The 043 Datasets

Fifteen expression, protein and morphology datasets: genotypes, agreement, co-location, and the design of one model over every gene-level readout

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October 19, 2026

Abstract

The 043 store holds 173,617 records of 15 yeast datasets that read mRNA, protein abundance or turnover, or CalMorph morphology, kept as served with no aggregation. Its genotypes fall into four classes, and today a model sees only one of them as intended: deletions are gene sets, isolates mark almost every gene as perturbed, segregants mark none, and the promoter library has no gene-level genotype. Each dataset is set against its own reproducibility. Kemmeren 2014's two cultures of 114 mutants give an own-culture ceiling of 0.729; it replaces the experiment 019 ceiling, which rested on Sameith 2015's single mutants, found here to be Kemmeren's own arrays. Two constructions of one Jackson 2020 genotype agree on the deletion effect at a median r of 0.129 against 0.095 for different genotypes; Hu 2007 released no replicate; no segregant appears in two panels. Between datasets, agreement is low (median per-genotype r from -0.006 to 0.208 apart from the Sameith copy), and morphology is the label most often shared with expression and protein. The plan trains every gene-level readout in one model and asks two engineering questions of it: held-out Kemmeren single deletions by gene, and all 72 Sameith double deletions against the additive null of their singles; the Muenzner proteome is a third test and Ohya morphology a probe of the frozen latent. The data as tensors, the input side (gene tokens, change-type marks and the perturbation operator) and the output side (readout sites, heads, tokens, losses and the evaluation protocol) follow, with the choices left to the owner collected in one table.

Section status: ✓ ready ■ author-reviewed, keep stable ✗ not done, agents may edit freely

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1 The 043 store ✗

Query. The store is one query over the served knowledge graph, KG 4.0 (release `2026.10.17-4295f7ca`), registered as the supported query `expression_proteome_morphology_043`. It takes every served yeast dataset whose readout is a vector of mRNA expression, protein abundance or turnover, or CalMorph morphology: 15 datasets, 173,805 served experiments. Five candidates were left out: Puddu 2019 holds a realized-genotype table rather than a cell-state readout, Ohnuki 2018 and 2022 share no strain with the other sets, Hackett 2020 carries an induction the graph processor would read as a deletion, and Messner 2021 carries no genotype perturbation.

The keep-the-data rule. The store holds one sample per served pair of experiment and experiment reference, with no unit converter, no deduplicator and no aggregation over genotypes. Two reasons were measured on the code: the mean deduplicator keys on the perturbed gene names alone, so it would average one deletion across Jackson 2020's eleven conditions, and it raises on four of the store's experiment types. Reconciling units and replicates is therefore a modeling decision, taken per batch, rather than a property of the store.

Size. The raw store holds 173,617 samples. The 188 served experiments it lacks name a gene outside the genome's 6,607-gene set and fail the deletion blocks' clause that every perturbed gene be in that set: Kemmeren 2, Hu 1, Nadal-Ribelles 70, Messner 75, Ohya 40 (slurm job 5855; `results/store_census.json`).

Terms. Table 1 fixes the words the rest of the text uses. Table 2 gives each dataset with the statistic that bounds what a model can reach on it.

Table 1: Controlled vocabulary, alphabetical.

Term	Meaning
Ceiling	The per-gene Pearson a perfect predictor of the true signal would score against one measurement: $\sqrt{\text{rel}_g}$, averaged over genes, with rel_g the share of variance across strains that is signal. It is a bound on a model, not a score of one.
Construction	One independently built strain of a genotype. Jackson 2020 built six per genotype, each with its own transcribed barcode, and the store keeps each as its own record.
Co-location	Two labels measured on the same genotype key, in any datasets.
Deletion-effect profile	Per gene, $\log_2((\text{CPM}+1)/(\text{CPM}_{\text{HO}}+1))$ for a construction against the HO-deletion control of the same condition.
Haplotype block	A run of markers on one chromosome over which a segregant's called parent and that call's posterior are constant; the genotype of a segregant is a list of blocks against two pinned parent assemblies.
Isolate	A natural strain; its genotype is thousands of sequence variants, gene presences and absences, and natural loss-of-function calls against the reference.
Pseudobulk	The UMI counts of several single cells summed per gene.
Segregant	A haploid meiotic product of a cross between two parent strains. Pooled segregants were genotyped by sequencing before profiling; de novo segregants are genotyped from each cell's own transcripts.
Test-retest	Two independent measurements of the same strains, correlated across strains per gene (r_g, so $\text{rel}_g = r_g$) or across genes per strain.

Table 2: The 15 datasets of the 043 store. Genotypes and records count the raw store; features per record is the median length of the stored label vector (genes, proteins or CalMorph features; one scalar for Vaishnav). Environments count distinct media, temperature and environmental perturbation; temperature is a typed gap where the paper states none. The last column gives each dataset's agreement with itself and the statistic used: a ceiling is the mean over genes of $\sqrt{\text{rel}_g}$ (Sec. 3.1); Jackson is the median Pearson between two constructions of one genotype against two of different genotypes, Nadal-Ribelles the median Pearson of one genotype across batches against different genotypes. Messner, Zelezniak, Ohya, Caudal and Nadal-Ribelles were measured in experiments 019 and 028, before this store.

Dataset	Genotype class	Readout, scale	Genotypes, n	Records, n	Features per record, median	Environments, n: name	Within-dataset agreement, statistic in cell
Kemmeren 2014	deletion	two-color array, log2 ratio to WT pool	1,482	1,482	6,169	1: SC, 30 C	0.73 own-culture ceiling, per gene (114 mutants with 2 cultures)
Sameith 2015 Sm	deletion	two-color array, log2 ratio to WT pool	82	82	6,169	1: SC, 30 C	none: the same arrays as Kemmeren (82 of 82 identical)
Sameith 2015 Dm	double deletion	two-color array, log2 ratio to WT pool	72	72	6,169	1: SC, 30 C	not measured
Hu 2007	deletion	error-model X score vs WT, signed	262	262	6,381	1: YPD, temperature a typed gap	not measurable: replicates averaged before release; Kemmeren × Hu r 0.11 is the bound
Jackson 2020	deletion, 6 construc-tions	scRNA-seq UMI counts, summed per construction	12	784	4,028	10: 9 media	construction pair r 0.13 vs 0.09 null, effect, floor 100
Nadal-Ribelles 2025	deletion	pseudobulk log2 fold change vs WT	3,059	6,118	5,866	2: 1 media	same genotype across batches r 0.043 vs 0.022 null
Messner 2023	deletion	SWATH protein abundance, linear	4,476	4,624	1,819	1: SM, 30 C	0.70 ceiling, 388 WT replicates
Zelezniak 2018	deletion	SWATH protein abundance, log	97	97	726	1: SM, 30 C	0.92 ceiling, per-strain replicate SE
Ohya 2005	deletion	CalMorph features, per feature	4,678	4,678	281	1: YPD, 25 C	0.61 ceiling, 122 WT replicates
Caudal 2024	isolate	RNA-seq TPM, linear	943	943	6,050	1: SC, 30 C	0.60 ceiling, 29 re-cultured isolates
Muenzner 2024 protein	isolate	SWATH protein abundance, linear	795	795	1,540	1: SM, 30 C	not measured
Muenzner 2024 turnover	isolate	SILAC degradation rate, per hour	55	55	1,055	1: SM + 80 mg/L [13C6/15N2] L-lysine, 30 C	not measured
Boocock 2025	segregant	scRNA-seq UMI counts, per cell or per segregant	79,516	79,516	962	2: SM, 30 C; 2x YNB + 2% glucose added to FACS sheath fluid, 30 C	no segregant in two panels; nearest-genotype deviation r 0.11 vs 0.07 random
N'Guessan 2025	segregant	scRNA-seq UMI counts, summed per segregant	2,951	2,951	2,592	1: YPD, temperature a typed gap	not measurable: cells summed per segregant, no replicate
Vaishnav 2022	promoter library	sort-seq expression level, one scalar	1	71,158	1	1: SC-Ura, 30 C	not measured
Total				173,617			

2 Genotype classes and how each enters a model ✗

The store holds four genotype classes, and the graph processors that turn a record into model input treat them differently. What follows is the behavior of the code at the 043 build (`torchcell/data/graph_processor.py`), measured on the processed samples.

Deletions. Kemmeren, Sameith, Hu, Jackson, Nadal-Ribelles, Messner, Zelezniak and Ohya: every record names one or two deleted genes (two for Sameith Dm). The genotype is the set of systematic names, and the `Perturbation` processor marks those nodes in `pert_mask`. Jackson 2020 adds a construction identity (plate well and barcode) below the gene name, so its 784 records hold 72 constructions of 12 genotypes.

Isolates. Caudal 2024 and the two Muenzner 2024 datasets: a natural isolate's genotype is its variant calls against S288C, five perturbation types with the counts of Table 3, between 195 and 7,025 perturbations per Caudal record. The `Perturbation` processor marks every gene with any of them, so a Caudal sample has 5,575 of 6,607 nodes in `pert_mask`: the mask then separates almost nothing, and a rule over perturbation types (for example loss of function and absence only) is needed before isolates are trained on.

Table 3: Perturbations of an isolate record, by type. Mean count per record over the records of each dataset (`results/store_census.json`). Sequence variants are named by the gene they fall in; natural loss of function, gene absence and gene presence are released calls; the few deletions are engineered markers.

Dataset	Records, n	Perturbations per record, min to max	Sequence variant, mean per record	Natural LoF, mean per record	Gene absence, mean per record	Gene presence, mean per record	Deletion, mean per record
Caudal 2024	943	195 to 7,025	5,047	383	135	84	0
Muenzner 2024 protein	795	0 to 7,025	5,017	377	133	84	0
Muenzner 2024 turnover	55	4,903 to 6,430	5,059	373	134	90	0

Segregants. Boocock 2025 and N'Guessan 2025: a `SegregantGenotype` names a cross, two parent strains pinned to sha256-verified assemblies, and a list of haplotype blocks, and it carries no perturbation list by design. Since the stage-2 fix the processors read a missing list as zero perturbed genes, so a segregant sample today has `pert_mask` all false: a model sees it as the reference strain. Using segregants needs a representation of the blocks; Sec. 5 names one.

Promoter library. Vaishnav 2022: each of 71,158 records is one 80 bp variable promoter insert driving a YFP reporter, read as one scalar. Every record names the same reporter, so the store holds one gene-name genotype for all of them; the perturbation sets separate them only by sequence content. No gene-level vector and no gene-set encoder applies.

3 Agreement within and between datasets ✗

3.1 Within a dataset ✗

A dataset's agreement with itself bounds both what a model can score on it and what any other dataset can agree with it. The new measurements are in `experiments/043-expression-proteome-morphology/scripts/within_dataset_ceilings.py` (slurm jobs 5893 for Kemmeren and Hu, 5895, 5896 and 5898 for Jackson, the segregants and the figures).

Jackson 2020. Each of the 784 records is one construction in one condition, a median of 23 cells and 43,851 UMIs (IQR 19,757 to 101,152); YPD and rapamycin reach a median of about 400,000 UMIs per construction, the other nine conditions 14,000 to 93,000. Within one condition, two constructions of the same genotype agree on their deletion-effect profiles only slightly better than two constructions of different genotypes (Table 4, Fig. 1a): a median r of 0.129 against 0.095 with every gene of at least 100 UMIs in the condition (1,778 same-genotype and 21,300 different-genotype pairs). The gap widens on the 660 deepest genes (0.244 against 0.103 at 1,000 UMIs) and when only genes covered in both constructions are kept, but that second rule scores fewer pairs, the deep ones. The different-genotype pairs are not near zero because both constructions are divided by the same HO reference. The Jackson ceiling for one construction's deletion effect is therefore low; most of a record's deviation from HO is count noise shared with any other construction in the flask.

Kemmeren 2014. GEO titles name the culture: `bck1-del-1-a` and `-1-b` are the dye-swapped arrays of culture 1. Of 1,480 mutants whose titles parse, 114 carry two cultures, 72 in the responsive series GSE42527 and 42 in the non-responsive series GSE42526; the other 1,366 have one (16 arrays with irregular titles are left out). Culture 1 against culture 2 of the same mutant gives a median per-strain r of 0.730, against 0.284 for culture 1 of one mutant and culture

Table 4: Jackson 2020, construction against construction within one condition. Pooled over the 11 conditions. Level profile: Pearson over genes of $\log_2(\mathrm{CPM}+1)$; deletion effect: of $\log_2((\mathrm{CPM}+1)/(\mathrm{CPM}_{\mathrm{HO}}+1))$ against the six HO constructions summed, HO pairs left out. A gene enters every pair of a condition when its UMIs summed over that condition reach the floor. Same genotype: two of the six constructions of one genotype; different: two constructions of different genotypes in the same flask.

Gene rule	Level profile r, median, same vs different genotype	Deletion-effect r, median, same vs different genotype	Pairs, n, same / different	Genes per pair, median
$\geq$ 10 UMIs summed over the condition	0.637 vs 0.620	0.184 vs 0.157	1,778 / 21,300	5,405
$\geq$ 100 UMIs summed over the condition	0.654 vs 0.612	0.129 vs 0.095	1,778 / 21,300	3,701
$\geq$ 1000 UMIs summed over the condition	0.919 vs 0.887	0.244 vs 0.103	1,778 / 21,300	660
$\geq$ 10 UMIs in both constructions	0.941 vs 0.925	0.406 vs 0.244	1,742 / 20,866	490
$\geq$ 100 UMIs in both constructions	0.968 vs 0.951	0.414 vs 0.164	1,191 / 12,337	34
$\geq$ 1000 UMIs in both constructions	0.997 vs 0.991	0.600 vs 0.112	271 / 2,924	0

2 of another (Table 5, Fig. 1b). Across the 114 mutants the median per-gene r is 0.612, so the own-culture ceiling is 0.729 by test-retest and 0.653 by decomposition (Fig. 1c). The two differ because the 114 paired mutants are enriched for strong responders, while the decomposition takes its signal variance from all 1,480. These numbers replace the void 0.775. The stored record averages both cultures where two exist; hypothesis, untested, that its ceiling is higher than the single-culture one.

Table 5: Kemmeren 2014, culture 1 against culture 2. A culture's profile is the mean over its dye-swapped arrays of the within-array $\log_2$(deletion/reference pool), computed by the loader's own functions from the GEO arrays of GSE42527 and GSE42526. The null pairs culture 1 of one mutant with culture 2 of another. Per-gene ceilings average over genes with at least three paired mutants; the decomposition takes the noise variance from the paired cultures and the total variance from the first culture of every mutant.

Kemmeren 2014 own-culture statistic	Value
Mutants with two cultures, n (responsive / non-responsive series)	114 of 1,480 (72 / 42)
Per-strain r across genes, median [IQR]	0.730 [0.602, 0.835]
Per-strain r, different mutants (null), median	0.284
Per-gene r across mutants, median [IQR]	0.612 [0.431, 0.754] (6,169 genes)
Ceiling, test-retest: mean over genes of $\sqrt{r_g}$	0.729
Ceiling, decomposition: mean over genes of $\sqrt{1-\sigma^2_{e,g}/\sigma^2_g}$	0.653
Void 019 cross-study ceiling (Sameith pair, identical arrays)	0.775

Hu 2007. No replicate statistic is computable. The paper says "Each experiment was independently carried out at least twice" (paper.md line 55), and the replicates were combined before release: the group "performed weighted duplicate averaging with custom Java software" (line 59). GEO GSE4654 holds one X score per strain; the per-array data went to the Longhorn Array Database and a lab site that was never archived. The only available bound is the agreement with Kemmeren, a median per-genotype r of 0.107 over 255 deletions and a per-gene r of 0.097 (Table 7). That agreement is a floor on Hu's reproducible signal; no ceiling is measured.

Boocock 2025 and N'Guessan 2025. Neither panel has a replicate: every pooled record sums the cells of one segregant (a median of 17 for Boocock Ap, 3 for N'Guessan), and the per-cell counts are not stored, so no split-half is possible (Table 6). No segregant appears in two panels: on a 5 kb grid of called parents the best concordance of any of the 393 pooled segregants with any of 27,744 de novo BY × RM cells is a median of 0.672, and with any of the 2,951 N'Guessan segregants 0.657, where unrelated segregants share half the genome and identity would be near 1; none reaches 0.95 (Fig. 2b). A weaker test still asks whether genotype carries expression across panels. For each pooled segregant, the 20 de novo cells nearest in genotype (mean concordance 0.647) agree with its deviation from the panel mean at a median r of 0.113, against 0.070 for 20 random cells; the paired difference is 0.040 (388 segregants, 63% positive, Wilcoxon $p = 1 \times 10^{-8}$). The single nearest N'Guessan segregant gives 0.101 against 0.060 (381 segregants, $p = 0.009$). At the panel level the summed BY × RM profiles agree at $r = 0.872$ between Boocock's pooled and de novo panels and 0.693 between Boocock's pooled panel and N'Guessan, which grew in YPD rather than minimal medium. Genotype posteriors differ by panel: a median 0% of markers of a pooled Boocock segregant and 1.7% of an N'Guessan one are called with posterior below 0.9, against 24 to 35% for a de novo cell.

3.2 Between datasets ✗

Records are matched by perturbed gene names and environment (isolates by strain name), and each pair is scored per shared genotype, across shared genes, and per shared gene, across shared genotypes (Table 7, Fig. 3; `experiments/043-expression-proteome-morphology/scripts/cross_dataset_correlation.py`, slurm jobs 5890 and 5892). Every pair compares log against log, so the base of a logarithm does not change a correlation.

Table 6: The segregant panels. A pooled record sums the cells matched to one previously genotyped segregant; a de novo record is one cell genotyped from its own transcripts. The posterior column is the marker-weighted fraction of a genotype whose called parent has posterior below 0.9.

Panel	Records, n	Cells, n	Cells per record, median	UMIs per record, median	Genes detected per record, median	Marker fraction with posterior < 0.9, median
Boocock Ap, pooled BY × RM	393	7,124	17	27,902	3,595	0.000
Boocock A, de novo BY × RM	27,744	27,744	1	4,017	1,228	0.236
Boocock B, de novo YJM145 × YPS163	44,784	44,784	1	2,130	859	0.352
Boocock 3004, de novo CBS2888 × YJM981	6,595	6,595	1	1,982	605	0.297
N'Guessan, pooled BY × RM	2,951	14,263	3	18,221	2,592	0.017

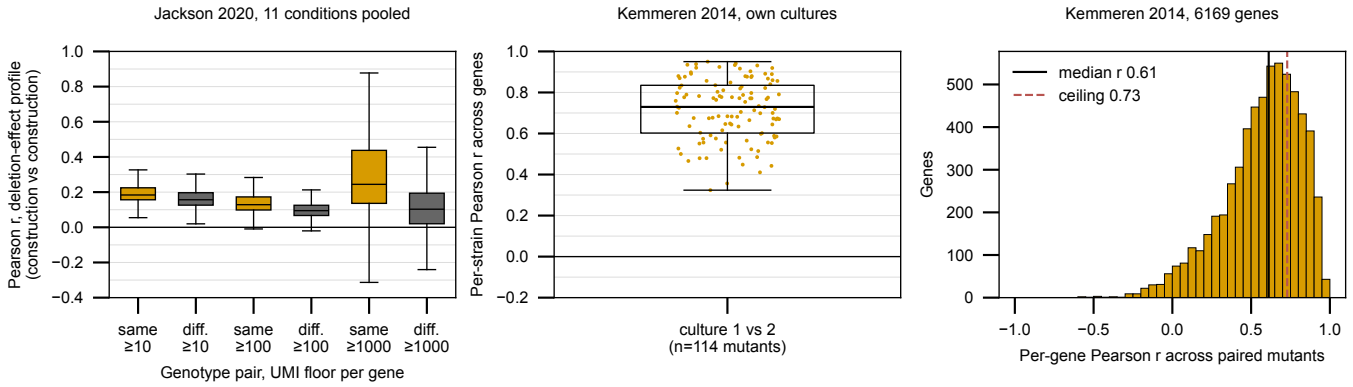


Figure 1: Two datasets measured against themselves. **a**, Jackson 2020: Pearson over genes of the deletion-effect profiles of two constructions in one condition, same genotype (orange) against different genotypes (gray), at three floors on a gene's UMIs summed over the condition; 1,778 and 21,300 pairs per floor, 11 conditions pooled, HO pairs left out. **b**, Kemmeren 2014: Pearson across genes between the two cultures of each of the 114 mutants profiled twice; culture 1 of one mutant against culture 2 of another gives a median of 0.284 (not drawn). **c**, per-gene Pearson across the 114 mutants for 6,169 genes, with the median r and the ceiling, the mean over genes of $\sqrt{\max(r_g, 0)}$.

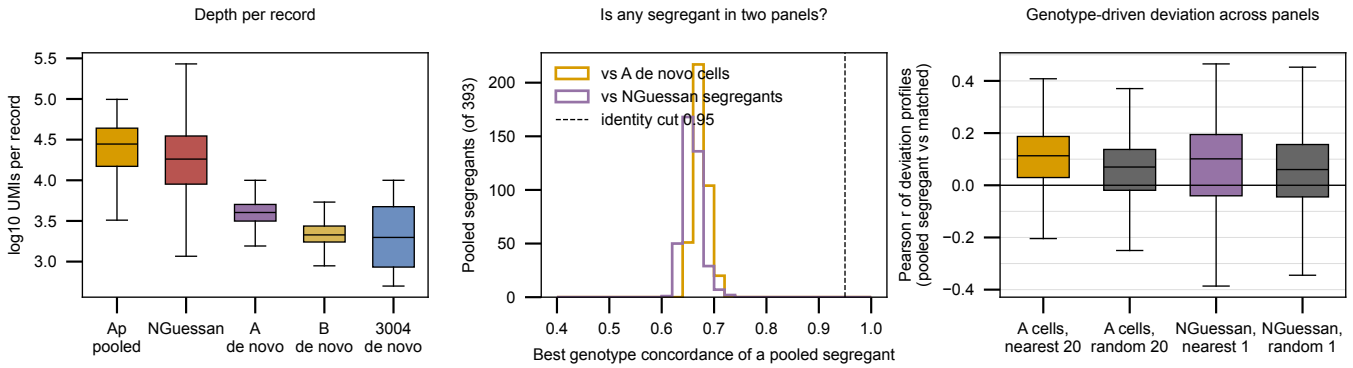


Figure 2: The segregant panels have no shared segregant and a weak shared genotype signal. **a**, UMIs per record by panel (pooled records sum a segregant's cells, de novo records are single cells). **b**, for each of the 393 pooled Boocock segregants, the best genotype concordance with any de novo cross-A cell (orange) or any N'Guessan segregant (purple), on a 5 kb grid of called parents; the dashed line marks 0.95. **c**, Pearson over genes between a pooled segregant's deviation from its panel and that of its 20 nearest de novo cells, 20 random cells, its nearest N'Guessan segregant, or a random one; genes with at least 10 UMIs on both sides.

Table 7: Agreement between datasets of the same readout type. Per genotype: Pearson and Spearman across the shared genes (at least 20). Per gene: Pearson across the shared genotypes (at least 5). Jackson is turned into a deletion effect against HO in the same environment; Messner and Zelezniak are log ratios to their record reference; Caudal and Muenzner are centered per gene over the isolates. The Jackson environment row pairs one genotype across two environments (605 genotype-environment pairs).

Pair	Shared genotypes, n	Per-genotype Pearson r, median [IQR]	Per-genotype Spearman, median	Per-gene Pearson r, median (genes, n)	Genes with per-gene r > 0.3, fraction
Kemmeren × Sameith Sm	82	1.000 [1.000, 1.000]	1.000	1.000 (6,169)	1.000
Kemmeren × Hu	255	0.107 [0.013, 0.184]	0.082	0.097 (5,803)	0.102
Kemmeren × Nadal, YPD	913	0.002 [−0.017, 0.022]	0.007	0.017 (5,891)	0.007
Kemmeren × Nadal, NaCl	920	−0.001 [−0.021, 0.022]	0.005	0.015 (5,896)	0.006
Kemmeren × Jackson, YPD 10 h	11	0.043 [0.026, 0.070]	0.047	0.035 (5,340)	0.247
Kemmeren × Jackson, YPD 4 h (a)	11	0.066 [0.027, 0.210]	0.030	0.118 (5,353)	0.321
Kemmeren × Jackson, YPD 4 h (b)	11	0.076 [0.036, 0.149]	0.045	0.052 (5,601)	0.253
Nadal YPD × Nadal NaCl	2,952	0.208 [0.168, 0.231]	0.043	0.183 (6,134)	0.050
Nadal duplicates, one condition	105	0.376 [0.323, 0.452]	0.178	0.062 (5,897)	0.089
Jackson, environment × environment	605	0.010 [−0.005, 0.029]	0.009	n/a	n/a
Messner × Zelezniak (protein)	89	0.049 [−0.016, 0.185]	0.050	0.085 (715)	0.067
Kemmeren mRNA × Messner protein	1,349	0.031 [−0.025, 0.112]	0.034	0.094 (1,832)	0.033
Caudal mRNA × Muenzner protein	759	0.208 [0.109, 0.307]	0.176	0.160 (1,572)	0.183
Muenzner turnover × abundance	55	−0.006 [−0.053, 0.047]	−0.003	−0.007 (1,120)	0.066

Sameith 2015 single mutants are Kemmeren 2014's arrays. All 82 Sameith Sm profiles are value-for-value identical to Kemmeren's for the same deletions (median per-genotype $r = 1.000$), and the GEO deposits show why. GSE42536 holds 144 Sameith single-mutant arrays: 115 are byte-identical re-deposits of GSE42527 or GSE42526 arrays, and the other 29 match a Kemmeren array exactly on the normalized Cy5 and Cy3 signal columns. Sameith's Methods defer to the earlier paper: "Full details of all gene expression profiling procedures have been described before [46]". The pair is therefore no replicate, and the experiment 019 expression ceiling built on it (test-retest 0.744 per strain, ceiling 0.77) is void; Table 5 replaces it. The 143 Sameith double-mutant arrays are new hybridizations.

Nadal-Ribelles 2025. Nadal-Ribelles agrees with Kemmeren at a median per-genotype r of 0.002 in YPD (913 deletions) and with itself not much better. Experiment 028 measured the same genotype across batches at a median r of 0.043 against 0.022 for different genotypes, and split halves at 0.080 against 0.051. The 0.376 of within-condition duplicates in Table 7 is Pearson driven by a few extreme log fold changes (the Spearman of the same pairs is 0.178); hypothesis, untested, that they come from genes with near-zero counts on one side.

Jackson 2020 across environments. One TF deletion's effect profile in one environment agrees with the same deletion's profile in another at a median r of 0.010 (605 pairs). The paper reports that TF deletions change far fewer genes than conditions do (96 differentially expressed genes in YPD, 160 in rapamycin), so a near-zero cross-environment agreement is what the paper leads one to expect; it does not show the data to be noise, and the within-condition construction pairs above measure the noise directly.

mRNA against protein. Deletions agree weakly: Kemmeren against Messner gives a median per-genotype r of 0.031 over 1,349 deletions and a per-gene r of 0.094 over 1,832 genes. Isolates agree more: Caudal against Muenzner gives 0.208 over 759 isolates and 0.160 per gene. Protein varies far more across isolates than across deletions (Fig. 4, left, x axes); hypothesis, untested, that this larger range is why the two readouts agree more there. Turnover against abundance is near zero (55 isolates).

Expression to morphology. A ridge from each gene-level matrix to the 281 CalMorph features of the shared Ohya genotypes (5-fold out-of-fold, one row-permutation null) is the one cross-label link that matters for a held-out probe (Table 8, Fig. 4, right). Kemmeren carries the most signal per genotype; Messner reaches nearly as far on three times the genotypes.

Table 8: Ridge from a gene-level matrix to CalMorph. Out-of-fold Pearson per CalMorph feature across held-out genotypes; inputs and targets z-scored and clipped at 8, penalty chosen on an inner split. The 116 moving features are the experiment 019 list (`calmorph_moving_features.json` on the multimodal-phenotype-retrospective worktree).

Input matrix	Genotypes shared with Ohya, n	Mean r over 281 features	Row-permutation null, mean r	Median r over 116 moving features
Kemmeren 2014	1,437	0.238	−0.005	0.348
Messner 2023	4,279	0.179	0.008	0.278
Hu 2007	259	0.114	−0.052	0.235
Sameith 2015 Sm	81	−0.014	−0.077	−0.085

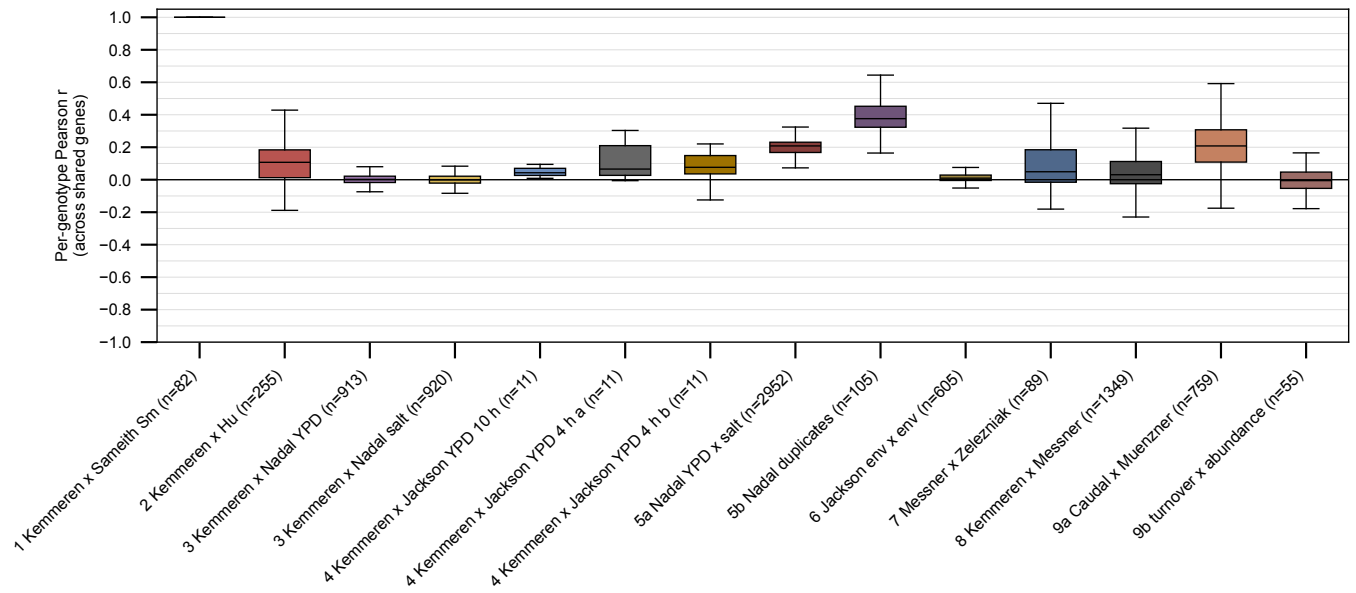


Figure 3: Agreement between datasets is low except for the Sameith copy. Per shared genotype, Pearson across the shared genes for every pair of Table 7; boxes are quartiles, whiskers 1.5 IQR, outliers not drawn.

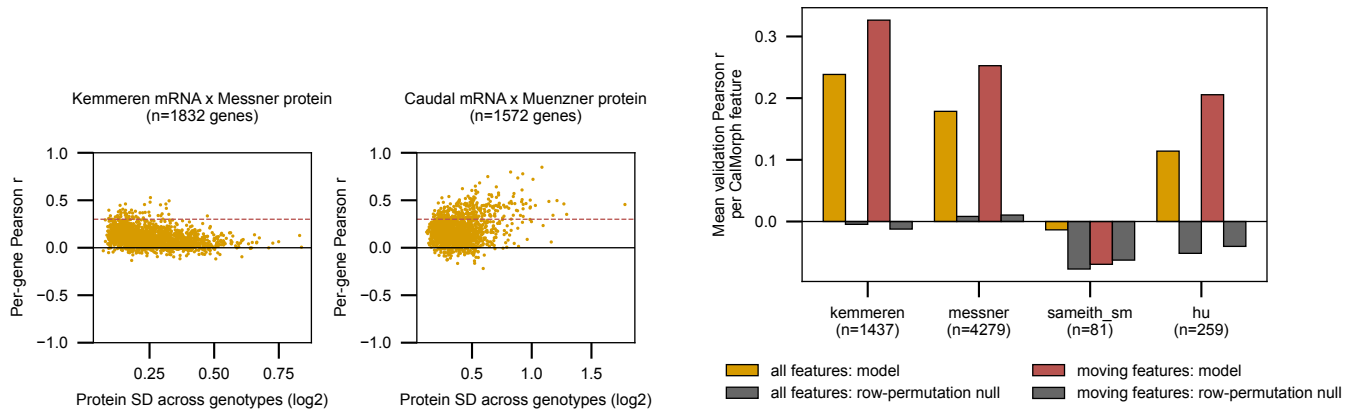


Figure 4: mRNA against protein, and expression to morphology. Left two panels: per-gene Pearson between mRNA and protein across the shared genotypes, against the protein's SD across those genotypes, for 1,832 genes over the deletions of Kemmeren and Messner and 1,572 genes over the isolates of Caudal and Muenzner; the dashed line marks $r = 0.3$. Right: mean out-of-fold Pearson per CalMorph feature of a ridge from each input matrix (Table 8), with its row-permutation null.

4 Which genotypes carry which labels ✗

A multitask model learns a shared representation only where one genotype carries more than one label. The census keys a genotype by its perturbed gene names, an isolate by its strain and a segregant by cross and identifier, and counts the genotypes each label and each pair of labels reach (Tables 9 and 10).

Table 9: Genotypes per label. Distinct genotype keys over every dataset carrying the label; Vaishnav is not keyed by gene and is left out.

Label	Genotypes, n
umi_count	82,479
protein_abundance	5,279
calmorph	4,678
expression_log2_ratio	3,685
expression_tpm	943
expression_score	262
degradation_rate	55

Table 10: Co-location. Genotypes carrying both labels, in any datasets. A label name can span datasets: `expression_log2_ratio` is Kemmeren, Sameith and Nadal-Ribelles; `umi_count` is Jackson, Boocock and N'Guessan; `protein_abundance` is Messner, Zelezniak and Muenzner.

Labels carried by the same genotype	Genotypes, n
expression_log2_ratio × calmorph	3,473
protein_abundance × calmorph	4,287
expression_log2_ratio × protein_abundance	3,317
expression_tpm × protein_abundance	759
umi_count × calmorph	12
expression_score × calmorph	259
isolates: Caudal × Muenzner protein	759
isolates: all three isolate datasets	55

Deletions. Morphology is the label most often shared. Of the 4,678 CalMorph genotypes, 4,287 also carry a protein profile and 3,473 an expression log ratio; the expression-by-protein overlap is 3,317 genotypes. That count includes Nadal-Ribelles, whose within-dataset agreement is near its null (Sec. 3.1); Kemmeren alone shares 1,349 genotypes with Messner. Kemmeren reaches Ohya on 1,437 genotypes and Messner on 4,279, the populations of the morphology ridge (Table 8).

Isolates. The isolate datasets are nested: all 55 turnover isolates are in the protein set and in Caudal's transcriptomes, and 759 of the 795 protein isolates have a Caudal transcriptome. No isolate carries a CalMorph profile.

Segregants and the rest. No segregant shares a key with any other dataset, and Boocock and N'Guessan share no segregant with each other (0 shared identifiers; Sec. 3.1 tests genotype identity rather than names). The 12 `umi_count` genotypes that carry morphology are Jackson's 11 transcription-factor deletions and the HO control.

5 The plan: every gene-level readout trained together ✗

Every gene-level readout of the store trains in one model, and the question it answers is an engineering one: does the model predict the transcriptome of a deletion it has not seen, and of a double deletion from its two measured singles? Table 11 gives the sets. An earlier version of this section proposed two separate training batches, deletions and then isolates with segregants; the owner superseded it on 2026-10-19.

Table 11: The sets of the plan. Records count the raw store (full pass of `data_shapes_043.py`, slurm job 5902); a split's count is strains held out, 10% of Kemmeren's 1,482 genes and a fifth of Muenzner's 795 isolates. Splits A and C are drawn from training records; split B is a dataset of its own and never trains.

Set	Datasets	Records or strains, n	Role
Training, every gene-level readout	Kemmeren 2014, Hu 2007, Jackson 2020, Nadal-Ribelles 2025, Messner 2023, Zelezniak 2018, Caudal 2024, Muenzner 2024 protein, Muenzner 2024 turnover, Boocock 2025, N'Guessan 2025, Vaishnav 2022	168,785	trained together; splits A and C are carved out of these records
Split A, Kemmeren by gene	Kemmeren 2014	148	10% of the deletion genes, out of every deletion dataset
Split B, Sameith doubles	Sameith 2015 Dm	72	every double; scored against the additive null of its two Kemmeren singles
Split C, Muenzner isolates	Muenzner 2024 protein (with Caudal 2024 and Muenzner 2024 turnover)	159	a fifth of the isolates, out of every readout of that isolate
Probe	Ohya 2005	4,678	ridge from the frozen latent to the CalMorph features; never trained
Dropped	Sameith 2015 Sm	82	the same arrays as Kemmeren (82 of 82)

Training. Twelve datasets train together: every dataset of the store except Sameith 2015 Sm, whose 82 profiles are Kemmeren's own arrays (82 of 82 identical), Sameith 2015 Dm, which is split B, and Ohya 2005, which is the probe.

Vaishnav 2022 stays in as the promoter signal: one sort-seq level per promoter insert, read off the strain-level summary and the promoter encoder of Sec. 7.2. Expression and protein readouts come off the perturbed graph: a gene's value is read from that gene's own perturbed node state, and the CLS state and the perturbed-gene summary z_S are the strain-level sites (Sec. 8.1). The low-agreement datasets train too, and their measured agreement is what the weighting decision (O5 in Table 12) and the arm without them read: Nadal-Ribelles agrees with itself across batches at a median r of 0.043 against 0.022 for different genotypes and with Kemmeren at 0.002; two Jackson 2020 constructions of one genotype agree at 0.129 against 0.095; Hu 2007 released no replicate and agrees with Kemmeren at 0.107 per genotype, on an X score that is not a log ratio; Zelezniak 2018 agrees with Messner at 0.049 per genotype over 715 proteins though its own ceiling is 0.916; Muenzner 2024 turnover agrees with abundance at -0.006 per isolate.

Split A, Kemmeren singles by gene. A tenth of Kemmeren's deletion genes, 148, is held out of every deletion dataset, so a held-out gene is never seen as a perturbation. The bound is Kemmeren's own-culture ceiling, 0.729 over 114 mutants with two cultures (Sec. 3.1); the 019 reference, by strain rather than by gene, is 0.180 ± 0.006 per gene over four runs.

Split B, the Sameith doubles. All 72 Sameith 2015 double deletions are held out and scored against their measured profiles and against the additive null of their two Kemmeren singles, the paper's own expected value. Sameith's Methods defer to Kemmeren's, so the doubles were profiled by the same procedure; no double has a replicate, and the Kemmeren ceiling of 0.729 is the bound for the same arrays.

Split C, the Muenzner proteome. A fifth of the 795 Muenzner 2024 proteome isolates, 159, is held out of every readout of those isolates (Caudal 2024 shares 759 of them). No Muenzner replicate ceiling is measured; the reference is how far another readout of the same isolate agrees, Caudal mRNA against Muenzner protein at a median per-isolate r of 0.208, against 0.031 for Kemmeren mRNA against Messner protein on deletions.

The morphology probe. Ohya 2005 CalMorph enters no training. Its 4,678 genotypes are run through the frozen model and a ridge reads the 116 moving features from the latent. A ridge on the measured inputs already reaches a mean r of 0.238 from Kemmeren expression (1,437 shared genotypes) and 0.179 from Messner protein (4,279) over all 281 features, against row-permutation nulls near zero, and Ohya's own ceiling is 0.611 over 122 wild-type replicates (Table 8). Sec. 8.5 gives the bars for the probe.

The genotypes the model has not yet read. Isolates and segregants need a gene-level genotype the `Perturbation` processor does not build: marking every isolate perturbation sets 5,575 of 6,607 nodes, and a segregant sets none. Sec. 7 gives the change-type marks that replace it, with the definite rule for segregants (a gene inherits the non-reference parent only inside a block called at posterior 0.9 or above). The posterior cut removes little on the pooled panels (a median of 0% of markers for Boocock pooled, 1.7% for N'Guessan) and a quarter to a third on Boocock's de novo cells, whose single cells carry medians of 2,000 to 4,000 UMIs; whether those cells train is decision P1. The measured genotype signal across segregant panels is small: nearest-genotype deviation r 0.113 against 0.070 for random pairs (Sec. 3.1).

Decisions. Table 12 collects every choice the design leaves to the owner, with its two options, the measured fact that bears on it and the proposal. The I rows come from the input side (Sec. 7.8), the O rows from the output side (Sec. 8.6), and the P rows from putting the plan together.

Table 12: Owner decisions, one checklist. Each row is one choice the design cannot make alone. The measured fact is the one its section cites; "open" marks a choice the section makes no proposal on. The last column is the section that carries the full argument.

Id	Decision	Option 1	Option 2	Measured fact	Proposed	Sec.
I1	Base gene token	learned identity table (the 019 model)	frozen sequence encoder (ProtT5, nucleotide-transformer windows, FUDT)	every encoder is already a per-gene row on main; the 019 campaign used none; no comparison on expression measured	open	7.8
I2	Absence and deletion	one change type	two change types	135 natural absences per Caudal isolate; the similarity is measurable after training as the cosine of the two embeddings	two	7.8
I3	Isolate marks in the mask	every type, $M \approx 5{,}565$	loss of function and absence, $M \approx 518$	operator cost scales with the largest M in a batch; within-isolate mRNA-protein agreement 0.208 does not separate the two	open	7.8
I4	Segregant gene membership	whole gene inside a block	gene midpoint inside a block	blocks end at marker positions and the crossover lies in the gap; straddling genes per segregant not counted	whole gene (the prototype)	7.8
I5	Chromosomal position	on for every genotype class	on only when segregants enter	no run bears on it; nearest-genotype deviation r 0.113 against 0.070 is the only segregant signal	open	7.4
I6	Strain tokens	extra keys of the operator	added to the CLS row	at $M = 1$ an extra key makes $M = 2$ and changes the softmax, which confounds a comparison with 019	extra keys (the prototype's layout)	7.8
I7	Operator normalization at large M	softmax, magnitude fixed	sigmoid gate, magnitude grows with M	operator A run at $M = 1$ to 3 only; sigmoid gate 0.200 against 0.210, one seed; nothing at M of 10 or more	open	7.8
I8	Accessory ORFs and the Vaishnav reporter	encode each as an extra token	report and skip	84 accessory presences per Caudal isolate; one reporter (YFP) for 71,158 Vaishnav records	skip accessory ORFs; encode the reporter, since Vaishnav trains	7.8
O1	Heads	joined by readout type, 6 heads with 15 per-dataset affines	one head per dataset, 15 heads	same-type agreement 0.002 to 0.107 (expression), 0.049 (protein); an affine cannot raise a correlation; no joint training scored	joined; split as the control arm	8.6
O2	Experiment-type token	flat id	ontology path	the served ontology is flat, 34 leaves under phenotype	flat now, path when the ontology carries one	8.6
O3	Where the tokens enter	at the head only	on the CLS row, or as an attended token	experiment 035: -0.007, -0.025, $+0.008$ against ridge for the three forms, every CI through zero	head only; attended token as an arm	8.6
O4	Loss	MSE	19-quantile pinball	019 morphology: MSE on z_S escaped 12 of 12 runs, pinball 8 of 10	MSE; pinball as an arm	8.6
O5	Dataset weighting	equal per dataset	by ceiling, or by record count	Boocock and Vaishnav hold 87% of the records; Kemmeren holds 1,482	equal, agreement-weighted for the low-agreement sets; record count as an arm	8.6
O6	Nadal-Ribelles sentinels	$\lvert \log_2 \mathrm{FC} \rvert > 20$ set to missing	keep the stored value	20.2% of cells are sentinels	missing	8.6
O7	Jackson target	deletion effect against the HO control of the same environment	absolute $\log_2(\mathrm{CPM} + 1)$ with the environment token	construction pairs agree at 0.129 against 0.095 on the effect; one genotype across environments at 0.010	the effect	8.6
O8	Split A leakage rule	held-out gene out of every deletion dataset	out of the expression datasets only	the arm without the natural-genome datasets measures the isolate and segregant side either way	every deletion dataset	8.6
O9	Split C size	a fifth of the isolates, 159	another fraction	no Muenzner ceiling; Caudal against Muenzner 0.208 per isolate is the reference	a fifth	8.6
O10	Budget and window	019's 500 epochs read over 400 to 499	a longer budget	019's 1,000-epoch runs scored lower over 900 to 999 than over 400 to 499 in 7 of 8	declare with the launch	8.6
P1	Boocock de novo cells	train all 79,123 single cells under the posterior rule	train the 393 pooled segregants only	de novo cells: median 2,000 to 4,000 UMIs, 24 to 35% of markers below posterior 0.9	open; the every-readout direction includes them	5
P2	Split A against split B	draw split A from Kemmeren genes outside the 82 Sameith singles	draw split A from all 1,482	split B's additive null needs both singles of each double in training; the overlap of a random draw is not counted	draw outside the Sameith singles	5

6 Data as tensors ✗

Figure 5 follows a record from the store to a batch, and Table 13 gives each dataset's shape from a full pass over the 173,617 records. A record is a pair of experiment and experiment reference; its genotype becomes a set of marked genes on the 6,607-gene index G of `gene_set.json`, its phenotype a vector y over G with `NaN` where a gene is not measured, and its environment, dataset and experiment type become integer tokens.

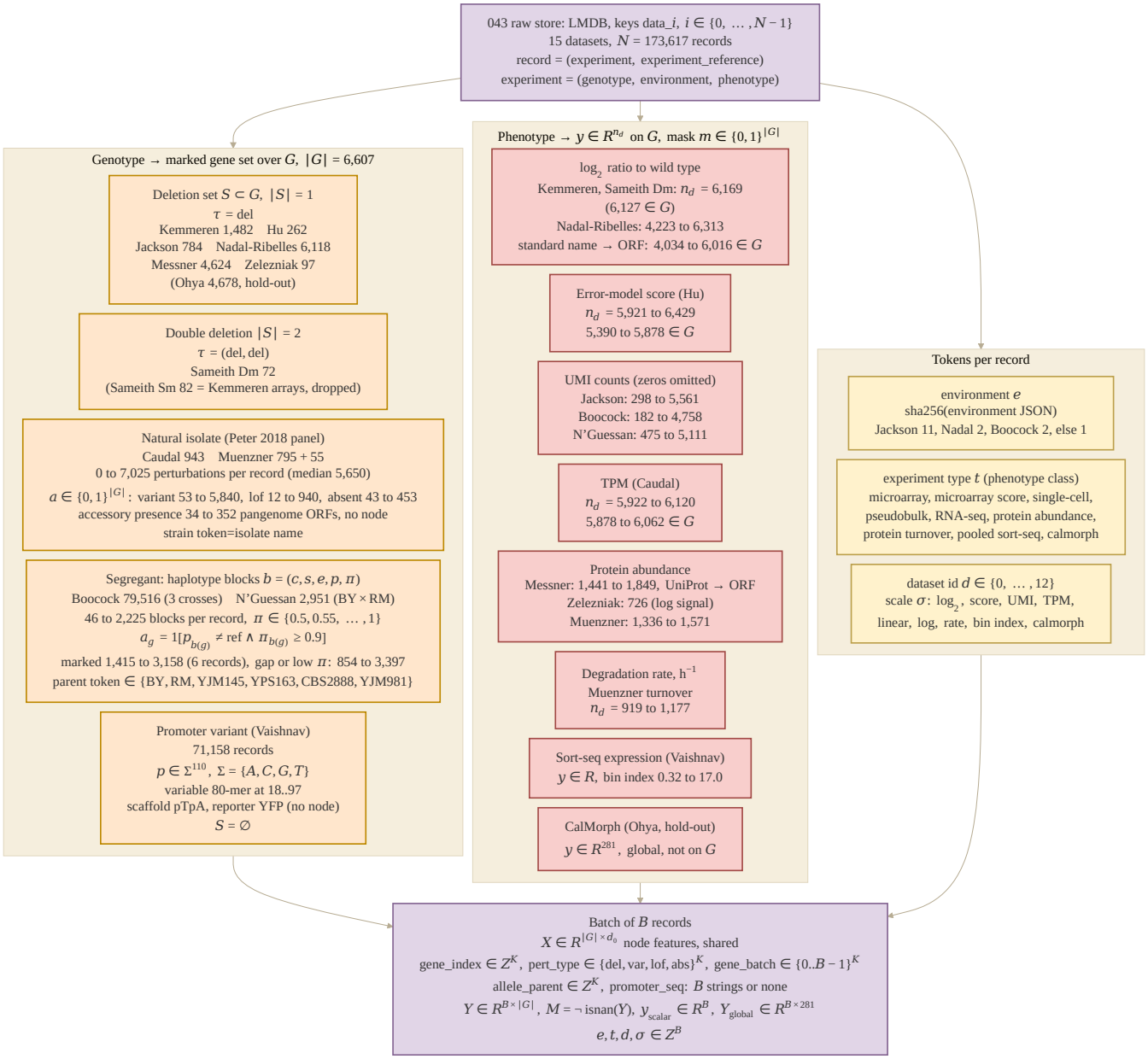


Figure 5: A record of the store becomes a marked gene set, a masked label vector and four tokens. Left, the five genotype classes and the counts each marks; middle, the nine released label scales with the number of label keys per record and how many of them fall on G; right, the environment, experiment-type and dataset tokens; bottom, the batch the collate function returns. Rendered 212 mm wide and placed at 180 mm, a scale of 0.85.

What the dataset module returns. Per sample, `dataset_043.py` returns the marked genes as indices on G (`gene_index`), a change type per marked gene (`pert_type`), the allele parent per marked gene for the segregants (`allele_parent`), the promoter sequence and scaffold for Vaishnav, the environment, dataset and experiment-type ids, the label scale (`y_scale`), the label $y \in \mathbb{R}^{6{,}607}$, one scalar for Vaishnav and the 281 CalMorph features for Ohya. The collate function concatenates the marked genes with a batch vector and stacks y into a $B \times 6{,}607$ tensor with its measured mask. A mixed batch of one record per dataset, 15 rows, carries 21,728 marked genes, 21,713 of them from the five isolate and segregant rows (the six-record debugger run).

Three shapes that bear on the design. The isolate and segregant rows are dense: a Caudal isolate marks 4,952 to 5,995 of 6,607 genes and a segregant 1,415 to 3,158 under the definite rule, against one for a deletion, so the perturbation

operator sees three orders of magnitude of marked-set size across datasets (counts on the debugger records). The definite rule leaves gaps: on the six debugger records 697 to 2,693 genes fall between blocks and 157 to 704 under a block below posterior 0.9. Two label-key maps run at read time: Nadal-Ribelles keys are standard names (1,441 of 6,796 on G by raw name, 6,325 after the map) and Messner keys are UniProt accessions (0 of 1,850 by raw name, 1,850 after). A key with no position on G is counted per sample and dropped from y, never guessed.

Table 13: Every dataset of the store as tensors. One row per dataset, from a full pass over the raw store (`data_shapes_043.py`, slurm job 5902). Perturbations per record count the released perturbation list (a segregant has none; its genotype is a list of haplotype blocks). Label values are the finite minimum and maximum over a sample of each dataset's values, on the released scale. Label keys on G count the keys that land on the 6,607-gene index after the read-time maps. Where a cell shows one number, minimum, median and maximum are equal; otherwise it reads median (minimum to maximum).

Dataset	Role in the plan	Records, n	Genotype class (records)	Perturbations per record	Label key, graph level	Label values, min to max	Label keys per record	Label keys on G per record	Environments, n
Kemmeren 2014	train; split A holds out 10% of genes	1,482	Genotype	1	`expression_log2_ratio`, node	−6.72 to 6.17	6,169	6,127	1
Sameith 2015 Sm	dropped: Kemmeren's arrays	82	Genotype	1	`expression_log2_ratio`, node	−4.31 to 5.23	6,169	6,127	1
Sameith 2015 Dm	split B: every double held out	72	Genotype	2	`expression_log2_ratio`, node	−6.25 to 4.8	6,169	6,127	1
Hu 2007	train	262	Genotype	1	`expression_score`, node	−8.35 to 8.05	6,381 (5,921 to 6,429)	5,832 (5,390 to 5,878)	1
Jackson 2020	train	784	Genotype	1	`umi_count`, node	1 to 21,655	4,028 (298 to 5,561)	4,009 (298 to 5,501)	11
Nadal-Ribelles 2025	train	6,118	Genotype	1	`expression_log2_ratio`, node	−33 to 32.9	5,866 (4,223 to 6,313)	5,665 (4,034 to 6,016)	2
Messner 2023	train	4,624	Genotype	1	`protein_abundance`, node	0.0313 to 393,221	1,819 (1,441 to 1,849)	1,819 (1,441 to 1,849)	1
Zelezniak 2018	train	97	Genotype	1	`protein_abundance`, node	1.62 to 11.2	726	726	1
Ohya 2005	probe only, frozen latent	4,678	Genotype	1	`calmorph`, global	0 to 56,974	281	none, features not genes	1
Caudal 2024	train; split C isolates held out	943	Genotype + natural isolate	5,662 (195 to 7,025)	`expression_tpm`, node	0 to 88,829	6,050 (5,922 to 6,120)	6,006 (5,878 to 6,062)	1
Muenzner 2024 protein	train; split C holds out a fifth of isolates	795	Genotype + natural isolate (794); Genotype (1)	5,628 (0 to 7,025)	`protein_abundance`, node	0.0906 to 181,277	1,540 (1,336 to 1,571)	1,540 (1,336 to 1,571)	1
Muenzner 2024 turnover	train; split C isolates held out	55	Genotype + natural isolate	5,662 (4,903 to 6,430)	`degradation_rate`, node	0.0169 to 2.44	1,055 (919 to 1,177)	1,055 (919 to 1,177)	1
Boocock 2025	train	79,516	SegregantGenotype	0	`umi_count`, node	1 to 2,301	962 (182 to 4,758)	962 (182 to 4,758)	2
N'Guessan 2025	train	2,951	SegregantGenotype	0	`umi_count`, node	1 to 12,339	2,592 (475 to 5,111)	2,591 (475 to 5,109)	1
Vaishnav 2022	train, promoter signal	71,158	Genotype	1	`expression_level`, node	0.321 to 17	1	none, one scalar	1
Total		173,617						6,607 genes on the index	

7 The input side: sequence, genotype and the perturbed cell graph ✗

Five genotype classes have to enter one model so that every gene-level readout trains together: engineered deletions and double deletions, natural isolates, segregants and promoter variants. What follows fixes the abstraction between the stored genotype and the perturbation operator: a **gene token** per reference gene from a swappable sequence encoder, a **change type** per gene per strain that says how the strain's copy of that gene differs from the reference, an optional **position** feature, and **strain tokens** for what is true of the strain rather than of one gene. The perturbation operator then reads marked genes exactly as it reads deleted genes today, with one addition that Sec. 7.6 names. Everything measured below is read off the 043 store census or the retrospective branch's model code; what is proposed is marked as such, and the decisions the design cannot make alone are collected in Sec. 7.8. Table 14 defines the symbols and Fig. 6 draws the path.

Table 14: Symbols of the input side, alphabetical. Widths are those of the 1,084,619-parameter expression model of experiment 019 (`conf/cgt_expr_012.yaml`: $d = 90$, $h = 9$ heads).

Symbol	Meaning
$a_i \in \mathcal{A}$	allele parent of gene i in a segregant: the parent whose block covers the gene; $\varnothing$ for every other gene and every other genotype class
B	strains in a batch
$c_i \in \mathcal{C}$	change type of gene i under the strain's genotype (Table 15); ref when the strain carries the reference copy
$\mathcal{C}$	the change-type vocabulary, seven members
d	width of a gene token and of a gene state, 90
$E \in \mathbb{R}^{(\lvert G\rvert+1)\times d}$	encoder output over the reference cell graph: row 0 the CLS state h, rows 1 to $\lvert G\rvert$ the wild-type gene states h_i
$f_\theta : \Sigma^* \to \mathbb{R}^{d_f}$	the sequence encoder, from a nucleotide or amino acid string to a vector; frozen or trained
$g_\phi : \Sigma^* \to \mathbb{R}^{d_g}$	the promoter encoder, from a promoter sequence to a vector
G	the reference gene set of the cell graph; $\lvert G\rvert = 6{,}607$
$H \in \mathbb{R}^{B\times\lvert G\rvert\times d}$	perturbed gene states, the operator's output (`H_genes_pert`)
h, $h^{\text{pert}} \in \mathbb{R}^d$	the CLS state before and after the operator
M	marked genes of a strain, $M = \sum_i m_i$; the operator's key count
$m \in \{0,1\}^{\lvert G\rvert}$	the mark: $m_i = \mathbf{1}[c_i \neq \text{ref}]$
$\mathbf{p}_t \in \mathbb{R}^d$	token of the t-th marked gene, the operator's key and value source
$\pi_i = (\chi_i, \rho_i)$	position of gene i: chromosome index $\chi_i \in \{0,\dots,16\}$ (0 the mitochondrion) and normalized coordinate $\rho_i = \frac{1}{2}(\text{start}_i + \text{end}_i)/L_{\chi_i}$
$s_i(\gamma) \in \Sigma^*$	realized sequence of gene i under genotype γ
$T \in \mathbb{R}^{\lvert\mathcal{C}\rvert\times d}$, $A \in \mathbb{R}^{\lvert\mathcal{A}\rvert\times d}$	learned change-type and allele-parent embeddings; $T_{\text{ref}} = A_\varnothing = 0$
$u_k \in \mathbb{R}^d$	the k-th strain token (parent identity, dataset, medium)
$X \in \mathbb{R}^{\lvert G\rvert\times d}$	reference gene tokens, one row per gene, shared by every strain
$\tilde{X} \in \mathbb{R}^{B\times\lvert G\rvert\times d}$	realized gene tokens of a batch, $\tilde{X}_{b,i} = X_i + T_{c_{b,i}} + A_{a_{b,i}}$
$z_S \in \mathbb{R}^d$	pool over a strain's marked rows of H (`pooled_perturbed`)

7.1 What enters the model today ✗

The 019 expression model builds its gene tokens from a learned table, not from sequence: `conf/train_cgt_multitask.yaml` sets `node_embeddings: []` and `learnable_embedding.enabled:true`, so X is an `nn.Embedding(6607,d)` indexed by gene order, and the CLS row is one learned vector shared by every strain. The encoder runs once per step on the wild-type graph; a strain enters only through the operator, whose token for a deleted gene is that gene's encoder state, $\mathbf{p}_t = h_{g(t)}$. The Methods write $\mathbf{p}_t = r(e_t) + \mathbf{t}_{\tau_t} + m_t\mathbf{w}$ with a perturbation-type embedding $\mathbf{t}_\tau$; the code implements neither the type term nor the magnitude term (perturbation-operator document, Table of symbols). Three consequences for the 043 classes are measured:

- A deletion and any other mark on the same gene produce the same key and the same value, since the token is the gene state alone.
- The `Perturbation` processor marks every perturbation of a record, so a Caudal isolate enters with 5,575 of 6,607 genes in `pert_mask` (Sec. 2), and under operator A their 5,575 values are averaged into one vector delivered to every gene.
- A segregant enters with `pert_mask` all false and is the reference strain.

The sequence encoders exist on `main` and are already keyed by gene: `NodeEmbeddingBuilder` serves ProtT5 on the protein (`prot_t5_xl_uniref50`), nucleotide-transformer windows of 5,979 bp around the ORF and of 1,003 bp upstream and 300 bp downstream, the fungal up-down transformer on the upstream and downstream flanks, CALM and codon

frequency on the coding sequence, and one-hot identity. Each is one frozen row per reference gene, concatenated into `cell_graph["gene"].x` and projected by the model's preprocessor. None of them has a per-strain version.

7.2 The gene token ✗

Definition. Gene i's token is the sequence encoder applied to its realized sequence under the genotype, $X_i(\gamma) = W_f\, f_\theta(s_i(\gamma))$, with $W_f \in \mathbb{R}^{d \times d_f}$ the projection to model width. For every unmarked gene $s_i(\gamma)$ is the reference sequence, so X_i is the reference token, computed once and shared by every strain. The encoder is the swappable part: f_θ may be any `NodeEmbeddingBuilder` product, a concatenation of several, a learned identity table (today's model, with f_θ the one-hot encoder and W_f the table), or a trained encoder.

Deleted gene: the row stays, marked. The deleted gene's row is neither removed from X nor zeroed. Removing it is the token-edit design, which costs one encoder pass per strain because the gene set then differs by strain (`notes-tex/perturbation-operator`, Sec. 4); the 019 configuration pays the encoder once per step and that is the dominant cost it was tuned around (`cgt_expr_012.yaml` header). Zeroing it before the encoder has the same cost and a second defect: a zero row through layer normalization is the normalized bias, one constant vector, so every deleted gene becomes the same token and the identity of what was deleted is lost before the first attention layer. The row therefore keeps its reference token and the deletion is carried by the change type, $c_i = \text{del}$, which the operator reads as a learned vector added to the token (Sec. 7.6). The gene's own readout entry is then predicted from a token that says "this gene, deleted", which is the information the deleted-gene flag $\delta_{i,t}$ of the perturbation-operator document supplies; the predicted $+0.036$ against a measured -2.42 for that entry is what the flag, and now the mark, exists to fix.

Variant gene: now the reference token plus a mark, later the variant sequence. For a natural isolate, every `SequenceVariantPerturbation` points at the isolate's allele through a sha256-pinned `sequence_ref` (`tc://genomes/peter2018_1011_assemblies/...#<gene>.fasta#<token>`), so $f_\theta(s_i^{\text{strain}})$ is computable for every variant gene. The census puts 5,047 variant genes per Caudal record over 943 records, 4,759,608 variant perturbations in Caudal and 3,988,281 in the Muenzner proteome; the two name the same isolates (759 shared), so the alleles overlap, and the number of distinct alleles is not in the census. Until an allele-keyed embedding store exists the variant token is $X_i + T_{\text{var}}$: the reference token plus one learned vector that says "this gene differs from the reference", the same vector for every variant gene. With the store it becomes $W_f f_\theta(s_i^{\text{strain}}) + T_{\text{var}}$, and the change type stays, because a model should know a token is non-reference without having to infer it from the feature. The two forms differ in what the model can learn: the first can learn that a gene is variant, the second which variant.

Loss of function and absence. A `NaturalLossOfFunctionPerturbation` is a released call on a gene (nonsense plus SIFT, or homozygous frameshift), 383 per Caudal record; a `NaturalGeneAbsencePerturbation` is a reference gene the isolate lacks, 135 per record. Both keep the reference row and carry their own change type. Most loss-of-function genes also carry a sequence variant; one gene carries one change type, resolved by precedence (Table 15).

Accessory gene present. A `NaturalGenePresencePerturbation` names a pangenome ORF with no row in G, 84 per Caudal record. It has no reference token and cannot be a row of X. The prototype reports every such mark per strain (`off_graph_marks`) and encodes nothing for it; including it means an extra token per accessory ORF from f_θ on its sequence, which the data supports (`sequence_ref`) and the prototype does not yet build.

Promoter variant. A Vaishnav record is one 80 bp variable region inside the constant `pTpA` scaffold driving YFP, 71,158 records under one gene-name genotype, so the only varying input is the promoter sequence p. Its token is the promoter encoder on p plus the reporter: $u_{\text{rep}} = W_g\, g_\phi(p) + T_{\text{prom}} + X_{\text{YFP}}$, where X_{YFP} is f_θ on the reporter's coding sequence (YFP is a gene addition, not a row of G, so this is an extra token like an accessory ORF). The readout is one scalar per record. Training g_ϕ on these 71,158 sequences is what the dataset is for; applying the trained g_ϕ to each native gene's upstream window (the 1,003 bp five-prime window already exists as a frozen nucleotide-transformer embedding) would give every reference gene a promoter term. Hypothesis (untested): a promoter term learned on the reporter library transfers to native promoters at all; the test is whether adding $g_\phi(\text{upstream}_i)$ to X_i moves the held-out Kemmeren score.

7.3 The change-type vocabulary ✗

Table 15 is the vocabulary and the map from every `perturbation_type` of the store onto it, with the count of marks per record. The mask is $m_i = \mathbf{1}[c_i \neq \text{ref}]$, so what the operator sees as perturbed is decided here and not by the processor's rule of marking everything. A training run may restrict further, for example marking only loss of function and absence for isolates; that is a mask over change types, $m_i = \mathbf{1}[c_i \in \mathcal{C}']$, and the count of distinct genes under each $\mathcal{C}'$ is not in the census yet (the 383 and 135 are counts of calls, and a gene can carry two calls).

Table 15: Change types, the store's perturbation types that map onto them, and marks per record. Counts are the census's perturbation counts divided by records (`results/store_census.json`); the isolate rows are means over records, the others exact. Precedence orders the types for a gene that carries several marks: the first in the order wins, so a gene that is gone outranks a call on its sequence, which outranks a present variant allele. Segregant marks are derived from haplotype blocks, not from a perturbation list.

Change type	Precedence	Perturbation types	Datasets	Marks per record
absent	1	`natural_gene_absence`	Caudal, Muenzner protein, Muenzner turnover	135, 133, 134
deletion	2	`kanmx_deletion`, `sga_kanmx_deletion`, `sga_natmx_deletion`, `barcoded_kanmx_deletion`, `marker_deletion`, `deletion`	Kemmeren, Hu, Jackson, Nadal-Ribelles, Messner, Zelezniak, Ohya, Sameith Sm (1 each); Sameith Dm (2); isolates (HO: 0.12, 0.11, 0.07)	1; 2; 0.12
loss of function	3	`natural_loss_of_function`	Caudal, Muenzner protein, Muenzner turnover	383, 377, 373
variant	4	`sequence_variant`; a segregant gene in a non-reference block	Caudal, Muenzner protein, Muenzner turnover; Boocock, N'Guessan	5,047, 5,017, 5,059; segregants not yet counted
present	5	`natural_gene_presence` (off graph)	Caudal, Muenzner protein, Muenzner turnover	84, 84, 90
promoter	6	`variable_promoter_reporter` (off graph)	Vaishnav	1
reference	implicit	none	every unmarked gene	$6{,}607 - M$

Segregants under the definite assumption. A `SegregantGenotype` is a cross, two parents pinned to assemblies and a list of `HaplotypeBlocks` ({chromosome, start, end, parent, posterior, n_markers} in reference coordinates). Gene i is marked $c_i = \text{var}$ with a_i the non-reference parent when one block of that parent with posterior ≥ 0.9 contains the whole gene, $\text{start}_b \leq \text{start}_i$ and $\text{end}_i \leq \text{end}_b$. A gene in a reference-parent block, in a block below the threshold, straddling a block edge or in the unassigned gap between two blocks stays reference: the crossover is unresolved by construction and nothing is guessed about it. The threshold removes little on the pooled panels (the median marker fraction below 0.9 is 0.000 for Boocock pooled and 0.017 for N'Guessan, Table 6) and much on the de novo cells (0.236 to 0.352). The parent token names the parent as the loader does: BY and RM for the pooled panels; YJM145, YPS163, CBS2888 and YJM981 for Boocock's other crosses. The loaders write chromosomes as `chrI` to `chrXVI` and the genome keys them as integers, so the gene table maps once. The probabilistic form, $m_i \in [0, 1]$ equal to the covering block's posterior, is the later step and changes nothing above except that the mask becomes a soft weight on the mark's embedding.

7.4 Chromosomal position ✗

Position is available for every reference gene from the genome object (`AnnotatedGene.chromosome`, `start`, `end`, `chromosome\_lengths`). As a feature it is $\pi_i = (\chi_i, \rho_i)$, entering the reference token as $X_i \leftarrow X_i + C_{\chi_i} + w_\rho\, \rho_i$ with $C \in \mathbb{R}^{17 \times d}$ a learned chromosome embedding and $w_\rho \in \mathbb{R}^d$ a learned direction. It is a flag in the prototype (`use_position`).

For including it now. A segregant's genotype is linkage: its marks come in contiguous runs along a chromosome, and the only way a model can treat two marked genes as one inherited block rather than two independent variants is to know they are adjacent. Aneuploidy in the isolate descriptor is per chromosome. Both are facts about position, not about sequence.

Against including it now. For the deletion datasets the graph is the wild type and position is a fixed per-gene attribute; a learned identity table already encodes any fixed per-gene attribute, so adding 17 one-hot plus one scalar to it adds nothing the table cannot represent. Position only adds information when the base token is a frozen sequence encoder with no identity channel, or when a gene is unseen in training. Nothing measured bears on whether linkage helps a segregant readout; the within-panel genotype signal is small (nearest-genotype deviation r 0.113 against 0.070, Sec. 3.1).

7.5 Strain tokens ✗

Some of what is true of a record is true of the strain, not of one gene: the parent identity of a segregant, the medium and temperature of the environment, the dataset and experiment type that fix the platform and the reference the label is relative to. The isolate clade is not on the graph: `NaturalIsolateDescriptor` carries isolate name, ploidy, zygosity, aneuploidy and manipulation, and no clade field, so "if released" resolves to not released. Each strain token is a learned vector u_k from a categorical vocabulary (prototype: `strain_token_vocab`, values such as `parent:RM`, `medium:YPD`, `dataset:kemmeren2014`).

Where they attach. Two placements are consistent with the single encode. (1) As extra keys and values of the

perturbation set, beside the marked genes: every gene attends to them through the same operator, and the CLS does too when `perturb_cls` is on. (2) Added to the CLS row before the operator, so they reach the genes only through the perturbed CLS. Adding them to every gene token is a third placement that makes the encoder input strain-dependent and so costs one encoder pass per strain; it is ruled out on the cost ground alone. Placement (1) is the one the prototype's output supports (`strain_tokens`, `strain_token_mask` in the operator's padded-key layout). The allele parent is the exception: it is a property of each marked gene, so it enters the gene token as A_{a_i}, and the strain token `parent:RM` says which parent is non-reference for the whole strain.

7.6 The perturbation operator with marks ✗

The operator is unchanged in form. For gene i and the strain's marked genes $t = 1, \dots, M$,

$$s_{i,t} = \frac{(h_i W_Q) \cdot (\mathbf{p}_t W_K)}{\sqrt{d_k}}, \qquad \Delta h_i = \sum_{t=1}^{M} \beta_{i,t}\, \mathbf{p}_t W_V, \qquad \hat{h}_i = \mathrm{LN}(h_i + \Delta h_i), \qquad h_i^{\mathrm{pert}} = \mathrm{LN}\big(\hat{h}_i + \mathrm{FFN}(\hat{h}_i)\big), \tag{1}$$

with $\beta_{i,t}$ the softmax over t (operator A), or the softmax over t and a null sink (operator C), as the perturbation-operator document defines them. What changes is the token:

$$\mathbf{p}_t = h_{g(t)} \quad \text{(code today)} \qquad \longrightarrow \qquad \mathbf{p}_t = h_{g(t)} + T_{c_{g(t)}} + A_{a_{g(t)}}, \tag{2}$$

which is the Methods' $\mathbf{t}_\tau$ term made concrete, with the allele parent beside it. At $T = A = 0$ it is the current operator exactly. Three things follow.

A variant and a deletion of the same gene are now different inputs. Under Eq. 2 the key and value of a marked gene depend on its change type, so the model can deliver a different vector to the genome for "*YFG1* deleted" and "*YFG1* carries the RM allele". Without it the two records are the same input with different labels, and the loss can only average them.

M goes from 1 to the hundreds or thousands. The deletion sets have $M = 1$ or 2; an isolate has $M \approx 518$ under loss of function plus absence and $M \approx 5{,}565$ under every type (sums of the Caudal means, before precedence); a BY $\times$ RM segregant has about half its genes in RM blocks if the parents segregate one to one (hypothesis; the count per segregant is read off the blocks and not yet computed). Operator A's softmax then averages M values into one vector whose magnitude does not grow with M, and the degeneracy at $M = 1$ disappears on its own at large M (the score chooses among thousands of keys). Operator B's sum would grow with M. Which normalization suits a strain with 500 marks is a decision (Sec. 7.8), and no run bears on it. The cost is arithmetic from the shapes: the batched operator pads to the largest M in the batch and scores $|G| \times M_{\max}$ pairs per head per strain, $6{,}607 \times 5{,}047 \approx 3.3 \times 10^7$ for a Caudal isolate under every type against 6,607 for a deletion, so mixing the two classes in one batch pads every deletion to the isolate's width.

The self entry changes meaning. For a deletion, gene $g(t)$'s own readout entry is the transcript that is gone (mean -2.42 in Kemmeren). For a variant it is the allele's own expression, which may be up, down or unchanged. The deleted-gene flag $\delta_{i,t}\mathbf{e}_{\mathrm{self}}$ therefore has to carry the change type, $\delta_{i,t}\,\mathbf{e}_{\mathrm{self},c_{g(t)}}$, or the model is told every mark silences its gene.

7.7 Shapes end to end ✗

For a batch of B strains over the reference gene set G,

$$X = W_f\, f_\theta\big(s(\gamma_{\mathrm{ref}})\big)\ [+ C_\chi + w_\rho \rho] \qquad \in \mathbb{R}^{|G| \times d}, \tag{3}$$

$$c \in \mathcal{C}^{B \times |G|}, \qquad m = \mathbf{1}[c \neq \mathrm{ref}] \in \{0,1\}^{B \times |G|}, \qquad a \in \mathcal{A}^{B \times |G|}, \tag{4}$$

$$\tilde{X}_{b,i} = X_i + T_{c_{b,i}} + A_{a_{b,i}} \qquad \in \mathbb{R}^{B \times |G| \times d}, \tag{5}$$

$$E = \mathrm{Enc}\left([\, x_{\mathrm{CLS}};\, X\,]\right) = [\, h;\, h_1; \dots; h_{|G|}\,] \qquad \in \mathbb{R}^{(|G|+1) \times d}, \tag{6}$$

$$\{\mathbf{p}_{b,t}\}_{t \le M_b} = \{\, h_i + T_{c_{b,i}} + A_{a_{b,i}} :\ m_{b,i} = 1 \,\} \ \cup\ \{u_{b,k}\}_k, \tag{7}$$

$$(h_b^{\mathrm{pert}},\, H_b) = \mathrm{Op}\left(h,\, \{h_i\},\, \{\mathbf{p}_{b,t}\}\right) \qquad \in \mathbb{R}^d \times \mathbb{R}^{|G| \times d}, \tag{8}$$

$$z_{S,b} = \mathrm{pool}\{H_{b,i} : m_{b,i} = 1\} \qquad \in \mathbb{R}^d. \tag{9}$$

The encoder in Eq. 6 runs once per batch on the reference tokens, as today; the strain enters at Eq. 7. Eq. 5 is what the prototype returns as `X_strain`; the operator consumes its marked rows through `perturbation_indices` and `perturbation_indices_batch`, the layout `EquivariantPerturbationTransform.forward` already takes, with

the type and parent terms added to the gathered keys. The readouts are unchanged: per-gene heads read H, the global head reads $[h^{\text{pert}} \,\|\, \text{mean}_i H_i \,\|\, z_S]$ (`GlobalHead(use_pert_summary)`), and z_S for a single deletion is the deleted gene's own perturbed token, for an isolate the pool over its marked genes.

The prototype, `torchcell/models/genotype_tokens_043.py`, implements Eqs. 3 to 5 and the strain tokens of Eq. 7: a `ChangeType` enum with the map of Table 15, a pydantic `GenotypeTokenSet` built from deletions, from a perturbation list or from haplotype blocks under the definite assumption, a `SequenceEncoder` protocol with a precomputed-table implementation and a `PromoterEncoder` protocol with no implementation, and a `GenotypeEncoder` module returning X, $\tilde{X}$, c, m, a, the flattened mark indices and the padded strain tokens. Its test (`tests/torchcell/models/test_genotype_tokens_043.py`, 4 tests) builds a deletion, an isolate-style and a segregant-style token set over 12 genes and checks every shape, the mask counts $(1, 4, 5)$, the precedence of loss of function over variant, the parent rows and that an accessory ORF is reported rather than dropped.

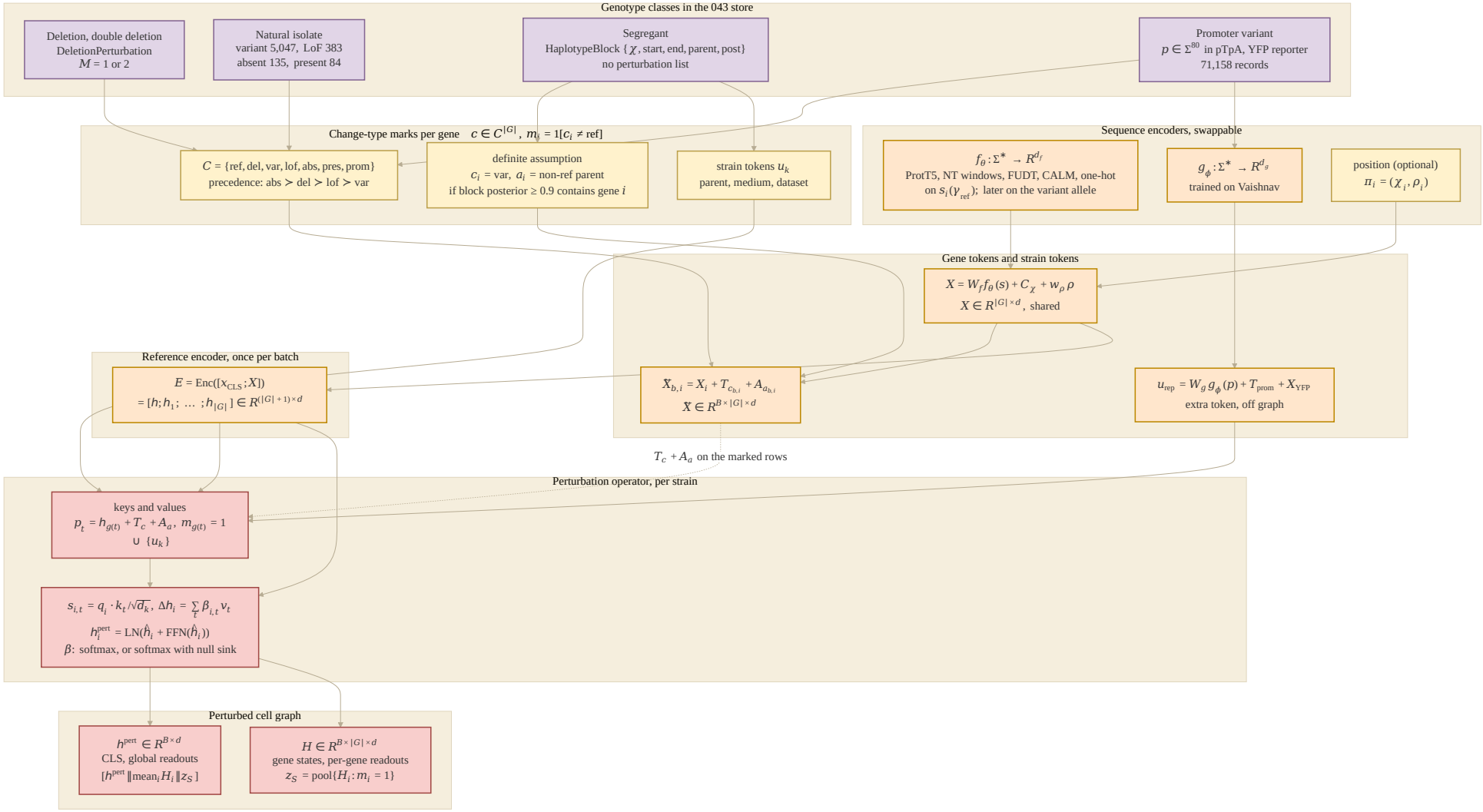


Figure 6: Every genotype class reaches the perturbation operator as marked rows of one shared set of gene tokens. Top, the four genotype classes of the store; second row, the change-type marks with their precedence, the definite segregant rule and the strain tokens, beside the swappable sequence, promoter and position encoders; third row, the reference gene tokens X, the realized tokens $\tilde{X}$ and the off-graph reporter token; the encoder runs once per batch on the reference tokens, and the operator takes the marked rows, with their change-type and allele-parent terms, as keys and values; bottom, the perturbed CLS state and gene states the readouts use. Placed on a rotated page at 262 mm, a scale of 1.24 of the 212 mm rendering.

7.8 Owner decisions ✗

Each item names the two options and the measured facts that bear on them; none is decided here.

1. **Base token: learned identity table or frozen sequence encoder.** Identity (today's model) can represent any fixed per-gene attribute and cannot say anything about an unseen allele. A frozen encoder (ProtT5, nucleotide-transformer windows, FUDT) gives a token for any sequence, including a variant allele and an accessory ORF, and is what makes "swap the encoder" meaningful. Facts: every encoder listed is already a per-gene row on `main`; the 019 campaign used none of them; nothing measured compares them on expression.

2. **Absence and deletion: one change type or two.** Both are the gene gone (`state="absent"`, SO deletion); provenance differs. One type hard-codes the transfer from 135 natural absences per isolate to the engineered single deletions; two types let the model learn the similarity, measurable as the cosine of T_{del} and T_{abs} after training. The vocabulary keeps two.

3. **Which isolate marks enter the mask.** Every type ($M \approx 5{,}565$, 5,575 of 6,607 genes marked in the processor today) or loss of function plus absence ($M \approx 518$). Facts: the operator cost scales with $M_{\max}$ of the batch; the within-isolate agreement of mRNA and protein (0.208 per isolate) is the only measured isolate signal and does not separate the options.

4. **Segregant gene membership: whole-gene containment or midpoint.** Containment leaves a gene straddling a block edge reference; midpoint assigns it. Facts: blocks end at marker positions and the crossover is in the gap; no count of straddling genes per segregant exists yet.

5. **Position: on for every class, or on only when segregants enter.** Facts in Sec. 7.4; no run.

6. **Strain tokens: extra keys of the operator, or added to the CLS row.** Both keep the single encode; the first lets every gene attend to the environment and parent directly, the second routes them through the CLS. Facts: operator A at $M = 1$ delivers one shared vector, so for a deletion in one medium placement (1) with one strain token makes $M = 2$ and un-degenerates the softmax as a side effect, which confounds a comparison against today's runs.

7. **Normalization at large M.** Softmax (average, magnitude fixed) or sigmoid gate (sum, magnitude grows with M). Facts: operator A has been run at $M = 1$ to 3; the sigmoid gate once at a resolving budget, 0.200 against 0.210, one seed (`notes-tex/perturbation-operator`, Sec. 3). Nothing at $M \geq 10$.

8. **Accessory ORFs and the reporter: encode as extra tokens or report and skip.** Both have sequences; neither has a row. Facts: 84 accessory presences per isolate; one reporter for Vaishnav. The prototype reports and skips.

8 The output side: readouts, heads, tokens, losses and the evaluation ✗

What follows is a design for the group to accept, change or reject. The owner's direction fixes three things: every gene-level readout is trained on; a readout is read off the perturbed graph, "from the node itself" for a gene-level value; and the question to answer is whether the model improves on held-out Kemmeren single deletions and on the Sameith double deletions, with Muenzner 2024 as a proteome test and Ohya 2005 CalMorph as a probe of the learned state and never a training target. Everything else below is a choice with its facts, Fig. 7 draws the path, and Sec. 8.6 lists the choices the owner has to make.

Table 16: Terms of this section, alphabetical. Terms of Table 1 are not repeated.

Term	Meaning
Additive null	For a double deletion xy and gene i, the single-mutant sum $M_{x,i} + M_{y,i}$, the expected value Sameith 2015 scores its epistasis against (Sec. 8.5). It needs no model.
Affine, per dataset	A scale a_k and shift b_k applied to a shared head's output for every record of dataset k; $a_k = 1$, $b_k = 0$ at the start. It moves a prediction between measurement scales and cannot change a correlation.
Escape	A run whose training Pearson, in eval mode, passes 0.3. In 019 the genotype-only morphology runs sat on a plateau at 0.04 to 0.12 that an escape leaves within about 50 epochs (retrospective, `11-inductive.tex`).
Head	The map from a readout site to a predicted value. Joined: one head per readout type, shared by its datasets. Split: one head per dataset.
Perturbed-gene summary z_S	The sum of the perturbed states of the strain's perturbed genes, $z_S = \sum_{g \in S} H_g$; for a single deletion it is the deleted gene's own token. The input of the 019 `M_zs` readout and of the trigenic-interaction head.
Readout site	A tensor of the perturbed graph a head reads: the perturbed CLS h, the summary z_S, or the gene states H.
Readout type	The physical quantity a head predicts: expression log ratio, absolute expression, expression score, protein abundance, protein turnover, promoter level.
Registered window	The epochs over which a run is scored, declared with the epoch budget before any run of the round is read; 019 used 500 epochs and epochs 400 to 499.
Token	A learned vector carried with a record and given to the model: the experiment type (ontology phenotype class), the dataset, and the environment.

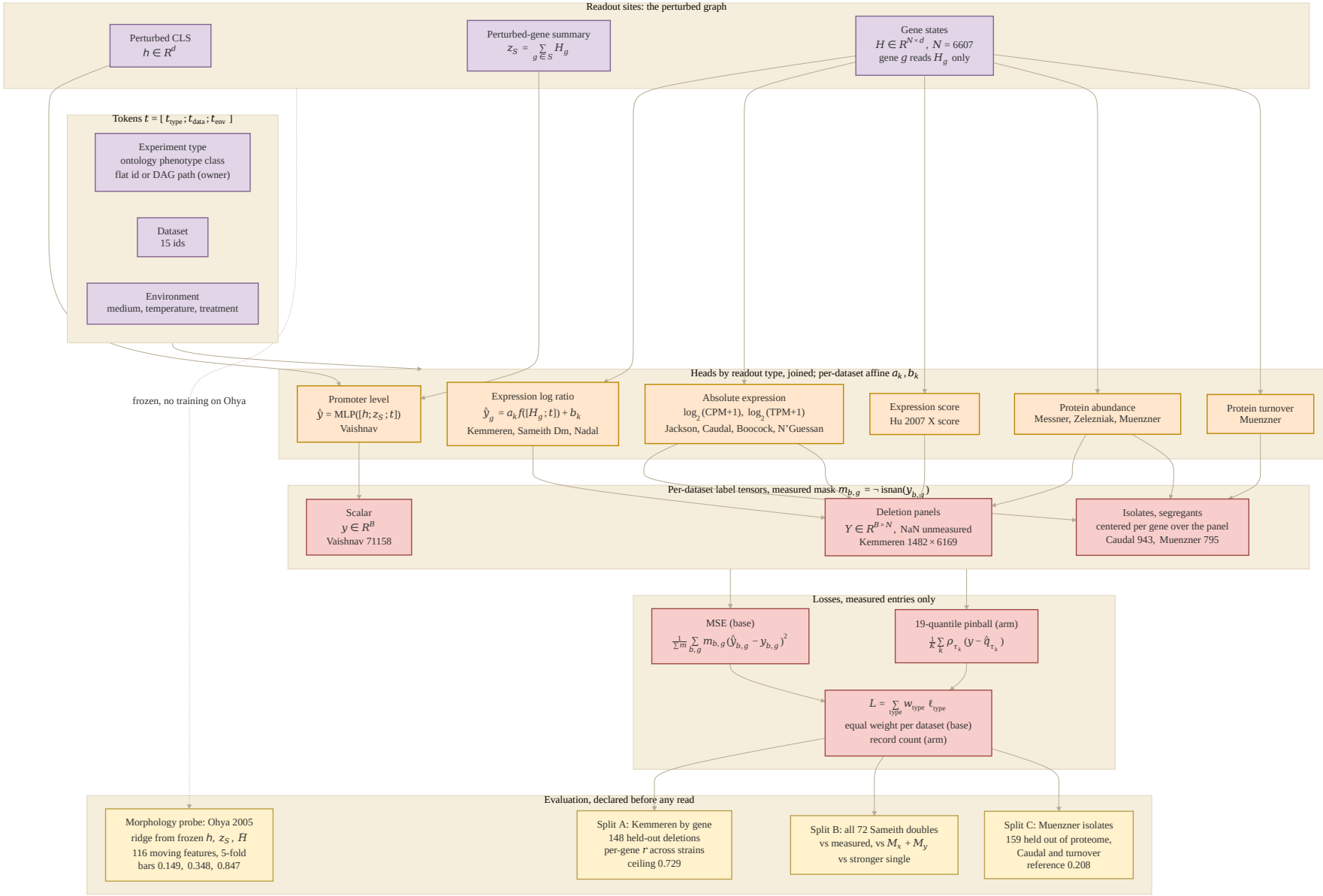


Figure 7: Gene-level readouts are read from each gene's own perturbed state through heads joined by readout type, and Ohya is read only from the frozen latent. Top, the three readout sites of the perturbed graph; left, the experiment-type, dataset and environment tokens; middle, the six heads with their per-dataset affines; then the label tensors with their measured mask and the losses over measured entries; bottom, the morphology probe and the three held-out splits. Placed on a rotated page at 241 mm, a scale of 1.14 of the 212 mm rendering.

8.1 Readout sites and the per-gene readout ✗

The trunk returns three tensors per record after the perturbation operator: the perturbed class token $h \in \mathbb{R}^d$, the perturbed gene states $H \in \mathbb{R}^{N \times d}$ over the $N = 6{,}607$ gene nodes, and the summary z_S. A gene-level value for gene g is read from that gene's own state,

$$\hat{y}_g = a_k \, f_\theta\big([\, H_g \,;\, t\,]\big) + b_k, \qquad t = [\, t_{\text{type}} \,;\, t_{\text{data}} \,;\, t_{\text{env}} \,], \tag{10}$$

with f_θ one head shared by every gene and every dataset of the readout type, t the record's tokens (Sec. 8.3) and (a_k, b_k) the affine of dataset k. The prediction for gene g depends on H_g and on nothing else of H; the strain reaches it through the trunk's attention and the perturbation operator, which is what "from the node itself" means in the equation. Four forms of f_θ are on the table.

- **Shared affine.** $f_\theta(x) = w^\top x + c$. The 019 head round measured the linear per-gene readout within noise of the two-layer one on Kemmeren expression (+0.002 over four seeds, `H_linear` against `H_ref`, `2-readouts.tex`), so the nonlinearity is not known to carry anything there.
- **Shared MLP with per-dataset affine** (Eq. 10; the prototype `PerGeneReadout`). The affine is where the datasets' scales part.
- **Shared MLP with the strain vector concatenated.** 019's `H_concat` feeds $[H_g; c]$ with c the strain context, and was the one arm above the reference in every seed, +0.0135 with a seed sd of 0.0026 on the difference (`2-readouts.tex`, Table `tab:readouts-head`). In Eq. 10 this is t extended by z_S.
- **Cross-attention head.** One learned query per output feature over every token (019 `M_xattn`). On morphology it escaped under the pinball (4 of 4, 0.121 ± 0.029) and not under MSE (1 of 4) (`11-inductive.tex`). It reads the whole token set, so it is not a readout from the node itself; it is the alternative if the per-gene form stalls.

A record-level value (Vaishnav 2022, one level per promoter insert) is read from $[h; z_S; t]$ through an MLP, the `GlobalHead` with `use_pert_summary` that gave the 019 morphology result (12 of 12 escapes with MSE, 0.149 ± 0.022 over the registered window, `11-inductive.tex`). The plain CLS escaped 0 of 8 under MSE at two learning rates, so h alone is not a readout site this design uses for a trained target; it remains a probe input (Sec. 8.5).

8.2 Joined or split heads ✗

Two facts bound the choice. First, the datasets report different quantities on different scales (Table 17): a two-color $\log_2$ ratio to a wild-type pool, an error-model X score, a pseudobulk $\log_2$ fold change with sentinels at ± 23 to ± 33 in 20.2% of cells (028, `2-readouts.tex`), UMI counts, TPM, a linear MaxLFQ abundance, a log SWATH signal, a degradation rate per hour, a sort-seq bin level. Second, datasets of one quantity agree weakly where they share a genotype (Table 7): Kemmeren against Hu 0.107, against Jackson 0.043 to 0.076, against Nadal-Ribelles 0.002 per genotype; Messner against Zelezniak 0.049; Kemmeren mRNA against Messner protein 0.031. An affine map preserves Pearson, so a per-dataset affine can reconcile the first fact and cannot reconcile the second: a head shared by two datasets that agree at 0.002 is asked to emit two unrelated vectors from the same H_g, told apart only by the tokens.

Joined by readout type. Six heads (expression log ratio; absolute expression; expression score; protein abundance; protein turnover; promoter level), each with one affine per dataset, 15 affines in all. The datasets that would share a head and the agreement that supports it: Kemmeren and Sameith Dm (the same arrays and procedures; Sameith's Methods defer to Kemmeren, Sec. 3.2; the Kemmeren ceiling 0.729 is the only within-type ceiling) and, under the same head by scale alone, Nadal-Ribelles at 0.002; Jackson, Boocock, N'Guessan and Caudal under absolute expression (no shared genotype among them; Boocock and N'Guessan share no segregant, Sec. 3.1); Messner, Zelezniak and Muenzner under protein abundance (Messner against Zelezniak 0.049; Muenzner shares no genotype with either). What a joined head buys is the transfer of the readout function across datasets, which is the premise of the exercise and is unmeasured; what it risks is that a large low-agreement dataset pulls the shared function away from the one with a ceiling.

Split by dataset. Fifteen heads, no sharing below the trunk. Datasets then meet only in H, which is what every 019 round did with two array datasets (Kemmeren and Sameith) that are the same arrays. Nothing is known about the trunk's transfer between readout types from this campaign: proteome from genotype reached 0.085 ± 0.008 (v16, 12 runs) and expression 0.180 ± 0.006 (v12, 4 runs) in separate trainings (`10-data-review.tex`); a joint training of both was not scored on both.

The proposal. Joined by type with per-dataset affines, under the dataset token, with the two low-agreement expression datasets (Nadal-Ribelles, Jackson) and Hu switched out as an arm (Sec. 8.5). The split design is the control arm of the same round, not a later step; the paired difference on the Kemmeren held-out score is the answer. Hypothesis, untested: the affine will absorb the scale and the dataset token will absorb the dataset's mean profile, leaving the shared head the per-gene response, which is what transfers.

Table 17: Readout type and target transform per dataset. The transform column is what the loss sees; "as released" means the stored value. The transforms for Jackson, Caudal, Messner, Zelezniak and Muenzner are the ones the cross-dataset analysis used (Table 7); the Nadal-Ribelles and Vaishnav rows are proposals.

Dataset	Readout type (head)	Released quantity, scale	Target transform
Kemmeren 2014; Sameith 2015 Dm	expression log ratio	two-color array, $\log_2$ (deletion / WT pool)	as released
Nadal-Ribelles 2025	expression log ratio	pseudobulk $\log_2$ fold change vs WT	as released, with $\lvert\cdot\rvert > 20$ set to missing (the zero-mean sentinel; 20.2% of cells)
Hu 2007	expression score	error-model X score vs BY4741, signed	as released; its own head, since an X score is not a log ratio
Jackson 2020	absolute expression	pseudobulk UMI counts, zeros omitted	$\log_2((\mathrm{CPM}+1)/(\mathrm{CPM}_{\mathrm{HO}}+1))$ against the HO deletion of the same environment, or $\log_2(\mathrm{CPM}+1)$ with the environment token carrying the condition (owner decision)
Boocock 2025; N'Guessan 2025	absolute expression	UMI counts per segregant	$\log_2(\mathrm{CPM}+1)$, centered per gene within the panel
Caudal 2024	absolute expression	RNA-seq TPM, linear	$\log_2(\mathrm{TPM}+1)$, centered per gene over the isolates
Messner 2023	protein abundance	SWATH MaxLFQ, linear	$\log_2$(strain / record reference)
Zelezniak 2018	protein abundance	SWATH log signal (SVA)	strain minus record reference
Muenzner 2024 protein	protein abundance	SWATH MaxLFQ precursor quantity, linear	$\log_2$, centered per protein over the isolates
Muenzner 2024 turnover	protein turnover	dynamic SILAC degradation rate, per hour	$\log_2$, centered per protein over the isolates
Vaishnav 2022	promoter level	sort-seq expression level, one scalar	standardized over the library

8.3 The experiment-type, dataset and environment tokens ✗

Three learned vectors travel with every record. The **experiment-type token** is the record's ontology phenotype class: `MicroarrayExpressionPhenotype` (Kemmeren, Sameith), `MicroarrayExpressionScorePhenotype` (Hu), `SingleCellExpressionPhenotype` (Jackson, Boocock, N'Guessan), `PseudobulkExpressionPhenotype` (Nadal-Ribelles), `RNASeqExpressionPhenotype` (Caudal), `ProteinAbundancePhenotype` (Messner, Zelezniak, Muenzner), `ProteinTurnoverPhenotype` (Muenzner) and `PooledSortSeqExpressionPhenotype` (Vaishnav), the classes of Table 2's rows. The **dataset token** is the dataset's id. The **environment token** is the record's medium, temperature and environmental perturbation, which separates Jackson's eleven conditions and Nadal-Ribelles' two.

The precedent. Experiment 035 gave the cell graph transformer a compound token z_E from a fingerprint MLP and asked whether it predicts Vanacloig 2022 for an unseen compound. Three entry points were measured on 123 compound-evaluations (`experiments.035`, rounds 9 and 10): a bilinear factor at the head (-0.007 paired against ridge, 95% CI -0.028 to $+0.015$), a gate on every gene's state (-0.025), and a token the genes attend to for one further layer ($+0.008$, CI -0.026 to $+0.045$; $+0.015$ against the head form, CI through zero). The ablation was the arm with no compound input (`cgt_genes`), whose centered score is zero by construction. The same three entry points and the same ablation apply here.

Where the tokens enter. (i) At the head only: t concatenated to H_g in Eq. 10 and to $[h; z_S]$ for the record-level head. Cheapest, and the 035 bilinear form's analog. (ii) On the CLS row: t added to the class token before the trunk, so every gene can attend to it. (iii) As a token the genes attend to in one further layer after the perturbation operator, the 035 environment-encoder form, the only one of the three with a positive paired mean against ridge there. The ablation arm carries no token, and a second ablation carries the dataset token only, which says whether the experiment type adds anything over the dataset id (hypothesis, untested: with 15 datasets and 8 classes the two are nearly collinear, and the type token earns its place only on a dataset unseen in training).

Flat id or ontology path. A flat token is one row per class. A path token sums one row per node of the class's ontology path, so `MicroarrayExpressionPhenotype` and `RNASeqExpressionPhenotype` share an expression row and differ in their own (prototype `ExperimentTypeEmbedding`, `mode="dag"`). The fact that decides it: the served graph has no such path. Every phenotype class is a direct child of `phenotype` in `biocypher/config/torchcell_schema_config.yaml` (the `phenotype member of` edge lists all 34 leaves), and in the pydantic schema only the two microarray classes share a parent below `Phenotype` (`TwoChannelMicroarrayExpressionPhenotype`). A path token therefore needs a grouping written on the modeling side (expression, protein, morphology, promoter), or it waits for the ontology unification to supply one. That is an owner decision (Sec. 8.6); the flat token needs nothing new.

8.4 Losses ✗

Each readout type has one loss over the measured entries of its records only: a missing gene is `NaN` in the target tensor, contributes no value and no gradient, and a type with nothing measured in a batch returns a connected zero (prototype `MaskedTypedLoss`, the convention of `masked_mean` in `torchcell/losses/distributional.py`). Three choices remain.

MSE or pinball. 019 trained expression with the 19-quantile pinball and read the median; on morphology, MSE on `M_zs` escaped 12 of 12 runs by epoch 410 with the smallest spread of any cell, the pinball on the same readout 8 of 10 and late (epochs 230 to 770), and MSE on the CLS 0 of 8 (`11-inductive.tex`). The loss and the readout interact, so neither is chosen for the per-gene heads by that result; the design runs MSE as the default and the pinball as an arm, both on the per-gene form. The Pearson loss and ListMLE are not arms: five of six batch-32 Pearson runs collapsed to a constant output and the anchored ListMLE runs sat on the mean by epoch 250 (`2-readouts.tex`).

Weighting across datasets. By record the store is 173,617 samples, of which Boocock holds 79,516 and Vaishnav 71,158 (Table 2), 87% of the records in two datasets with no gene-level deletion genotype; Kemmeren is 1,482. A loss averaged over measured entries weights datasets by records times features, so the deletion panels would be a few percent of the gradient. Three rules: equal weight per dataset (each dataset's mean loss summed); weight by the ceiling, so Kemmeren (0.729), Messner (0.696) and Zelezniak (0.916) count more than Nadal-Ribelles (0.043 against 0.022 across batches) and Jackson (0.129 against 0.095); or by record count. The proposal is equal weight per dataset with the low-agreement datasets down-weighted by their agreement, and the record-count rule as the arm that shows what the segregant panels do to the deletion score.

Scale. Every linear quantity is logged before the loss (Table 17); every pair in Table 7 was compared log against log, and a correlation does not depend on the base. Centering per gene over a panel (Caudal, Muenzner, the segregants) is what the cross-dataset analysis did and is the only way those panels have a zero; it means their head predicts a deviation from the panel mean, the same object a deletion dataset's log ratio is.

8.5 The evaluation protocol ✗

Three held-out sets, one probe, four arms and a scoring rule, declared before any run is read (Table 18).

Split A, Kemmeren by gene. A fraction of Kemmeren's 1,482 deletion genotypes, 10% or 148 genes, is held out by gene: no record in any deletion dataset whose perturbation set contains a held-out gene is trained on (Kemmeren, Hu, Jackson, Nadal-Ribelles, Messner, Zelezniak, Ohya), so the gene is unseen as a perturbation on any readout. A held-out gene still appears in the isolates and segregants as a sequence variant, an absence or an inherited block; those are not deletions and are not held out, and an arm without the natural-genome datasets (below) measures what that leakage, if it is one, is worth. The score is the per-gene Pearson across the 148 held-out strains for each of the 6,169 genes (mean and median over genes) and the per-strain Pearson across genes (median over strains), both against the own-culture ceiling 0.729 (test-retest, 114 mutants) and 0.653 (decomposition, Table 5). The 019 reference on Kemmeren plus Sameith is 0.180 ± 0.006 per gene over four runs and 0.233 for one 10,000-epoch run (`10-data-review.tex`); the 019 split was by strain with the same gene set, so it is a reference and not a paired bar. The Sameith Sm records are excluded from the store's training set outright: they are Kemmeren's arrays (82 of 82 identical).

Split B, every Sameith double. All 72 Sameith 2015 double deletions are held out; their constituent singles (82 distinct deletions, the Sameith Sm set) are in training as Kemmeren profiles, so the task is inference into the combinatorial space from measured singles. Each double is scored against its measured profile (per-strain Pearson over 6,169 genes; per-gene Pearson across the 72 doubles) and paired against two nulls that need no model: the additive null $M_x + M_y$ from the two Kemmeren singles, which is the paper's own expected value ("the deviation between the expression change observed in the double mutant $M_{x\Delta y\Delta i}$ and the expected expression change, given each single mutant $M_{x\Delta i} + M_{y\Delta i}$", paper.md line 39), and the stronger single alone. A third score restricts the genes to those where the double departs from the additive null by more than the paper's threshold, $|M_{xy,i} - (M_{x,i} + M_{y,i})| > \log_2 1.5$: on those genes the additive null is wrong by construction and the question is whether the model is less wrong. Sameith ran "four times from two independent cultures" per mutant (paper.md line 33) and released one profile, so no double has a ceiling of its own; the Kemmeren 0.729 is the bound for the same arrays and procedures.

Split C, Muenzner isolates. A fifth of the 795 Muenzner 2024 proteome isolates, 159, is held out by isolate, and the same isolates are held out from Caudal 2024 (759 shared) and from the turnover set (55 isolates), so a held-out strain is unseen on every readout. The score is the per-protein Pearson across the held-out isolates over 1,540 proteins and the per-isolate Pearson across proteins. No Muenzner replicate ceiling is measured (Table 2); the Caudal against Muenzner agreement, 0.208 per isolate and 0.160 per gene (Table 7), is the reference for what another readout of the same strain reaches, and the 019 proteome-from-genotype reading on Messner deletions, 0.085 ± 0.008, is the reference for the deletion head. Whether an isolate can enter the trunk at all is the input-side design's question; split C is declared so that the answer has a test ready.

The morphology probe. Ohya 2005 enters no training. For each of its 4,678 genotypes the frozen model produces h, z_S and the mean over genes of H; a ridge from each, and from their concatenation, to the 116 moving CalMorph features is fit 5-fold over genotypes with the penalty chosen on an inner split, inputs and targets z-scored and clipped at 8, and scored as the out-of-fold Pearson per feature across held-out genotypes, median over the 116 (the construction

of Table 8). The bars, in order: the 019 genotype-only model read through its own trained head, 0.149 ± 0.022; the sequence-embedding ridge 0.08 to 0.14 (`11-inductive.tex`); the Kemmeren-expression ridge 0.348 on the 1,437 shared genotypes and the Messner ridge 0.278 on 4,279 (Table 8), which read the strain's own measurement and are the transductive bars; and the ceiling 0.847 on the 116 features (019) or 0.611 over the 281 (122 wild-type replicates, Table 2). A probe at or above 0.348 from the genotype alone would say the latent carries what the measured expression carries; one near 0.149 says the probe is the trained head by another route.

Arms. Four arms in a 2×2: with and without the natural-genome datasets (Caudal, Muenzner protein and turnover, Boocock, N'Guessan), with and without Vaishnav; plus the split-head control and the no-token ablation on the full arm, and the deletion-only reference (Kemmeren and Messner; Sameith Dm is split B and does not train). Every arm is scored on splits A, B and the probe; split C exists only in the arms that train isolates. Three initializations and two split seeds per arm, as 019's v25 (four split seeds, three initializations) found the escape to be a property of the run and not the split.

Window and reads. The epoch budget and the window are declared with the launch; 019's rule was 500 epochs read over epochs 400 to 499, declared 2026-10-12, and its 1,000-epoch runs scored lower over 900 to 999 than over 400 to 499 in 7 of 8. A cell is read first by its escape fraction and second by its window mean; no maximum over epochs is reported. The test indices of every split are pinned to a file before training and the split-B and split-C scores are read once, at the end of the round.

Table 18: The evaluation protocol. Every bar is a measured number; a reference is a number from a different split or model and is marked so.

Split	Held out	n	Metric	Bar
A, Kemmeren by gene	10% of Kemmeren deletion genes; every deletion record naming one, in every deletion dataset	148 strains, 6,169 genes	per-gene Pearson across held-out strains (mean, median over genes); per-strain Pearson across genes (median)	ceiling 0.729 test-retest, 0.653 decomposition; reference 019 by-strain 0.180 ± 0.006
B, Sameith doubles	all 72 double deletions; the 82 constituent singles stay in training as Kemmeren records	72 strains, 6,169 genes	per-strain Pearson vs measured; paired difference vs the additive null $M_x + M_y$ and vs the stronger single; the same on genes with $\lvert M_{xy} - (M_x + M_y)\rvert > \log_2 1.5$	additive null (no model); ceiling 0.729 (same arrays, no double replicate)
C, Muenzner isolates	a fifth of the proteome isolates, the same strains out of Caudal and turnover	159 isolates, 1,540 proteins (turnover: those of its 55 among them)	per-protein Pearson across held-out isolates; per-isolate Pearson across proteins	no Muenzner ceiling measured; reference Caudal × Muenzner 0.208 per isolate, 0.160 per gene; reference 019 Messner 0.085 ± 0.008
Morphology probe	Ohya 2005, never trained; 5-fold over genotypes	4,678 genotypes, 116 moving features	ridge from frozen h, z_S, $\bar{H}$, their concatenation; out-of-fold Pearson per feature, median over 116	019 trained head 0.149 ± 0.022; sequence ridge 0.08 to 0.14; Kemmeren ridge 0.348 (1,437), Messner 0.278 (4,279), transductive; ceiling 0.847 (116) or 0.611 (281)

8.6 Owner decisions ✗

1. **Joined or split heads.** Joined by readout type with per-dataset affines (six heads, 15 affines), or one head per dataset (15 heads). Facts: same-type agreement is 0.002 to 0.107 for expression and 0.049 for protein (Table 7); an affine cannot raise a correlation; no joint training across types has been scored. Proposal: joined, with split as the control arm.

2. **Flat or path token for the experiment type.** The served ontology is flat (34 leaves under `phenotype`); a path needs a modeling-side grouping or the unification. Proposal: flat now, path when the ontology carries one.

3. **Where the tokens enter.** Head only, CLS row, or attended token (035's three forms; -0.007, -0.025, $+0.008$ against ridge, all CIs through zero). Proposal: head only as the base, attended token as the arm.

4. **Loss.** MSE or the 19-quantile pinball on the per-gene heads (019: MSE on z_S 12 of 12 escapes, pinball 8 of 10, on morphology only). Proposal: MSE base, pinball arm.

5. **Dataset weighting.** Equal per dataset, by ceiling, or by record count (two datasets hold 87% of the records). Proposal: equal per dataset, agreement-weighted for the low-agreement sets, record count as an arm.

6. **Nadal-Ribelles sentinels.** Treat $\lvert \log_2 \mathrm{FC}\rvert > 20$ as missing (20.2% of cells), or keep the stored value. Proposal: missing.

7. **Jackson target.** Deletion effect against the HO control of the same environment, or the absolute $\log_2(\mathrm{CPM}+1)$ with the environment token carrying the condition. Facts: construction pairs agree at 0.129 against 0.095 on the effect, and one genotype across environments at 0.010. Proposal: the effect.

8. **Split A leakage rule.** Hold the gene out of every deletion dataset (proposed) or out of the expression datasets only; the natural-genome arm measures the isolate and segregant side either way.

9. **Split C size and existence.** A fifth of the isolates held out, contingent on the input-side design admitting isolates at all.

10. **Budget and window.** Epochs and window to declare with the launch; 019's 500 and 400 to 499 are the precedent.